

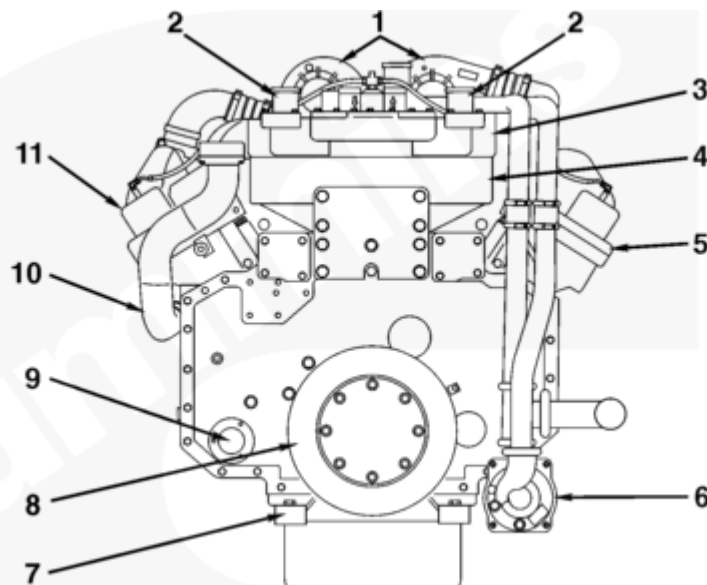
Trainings ▾

Engine Views

The illustrations show the locations of the major external components, filters, and other service and maintenance points. The location of certain external components will vary by engine model and engine configuration. Examples of some configurations are shown.

QSK45 Engines

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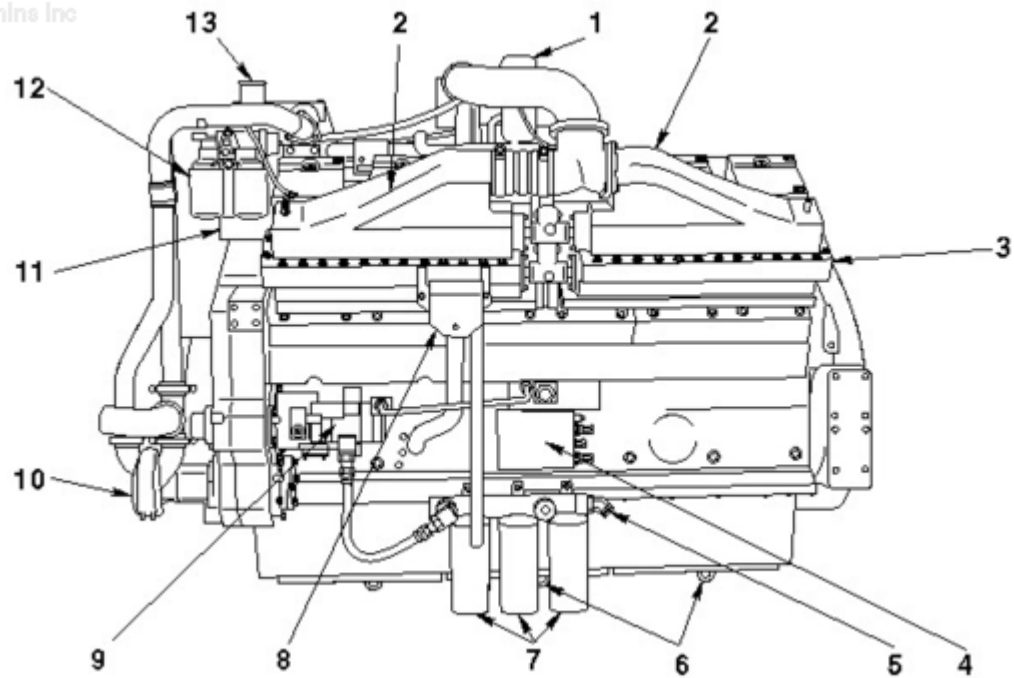
QSK45 - Front View

Note : Some engine configurations have the fuel return lines positioned at the front of the engine.

1. Turbochargers
2. Engine water outlet connections
3. Thermostat housing
4. Thermostat housing support
5. Left bank aftercooler assembly
6. LTA water pump
7. Front engine mount
8. Vibration damper
9. Alternator drive
10. Water bypass tube

11. Right bank aftercooler assembly.

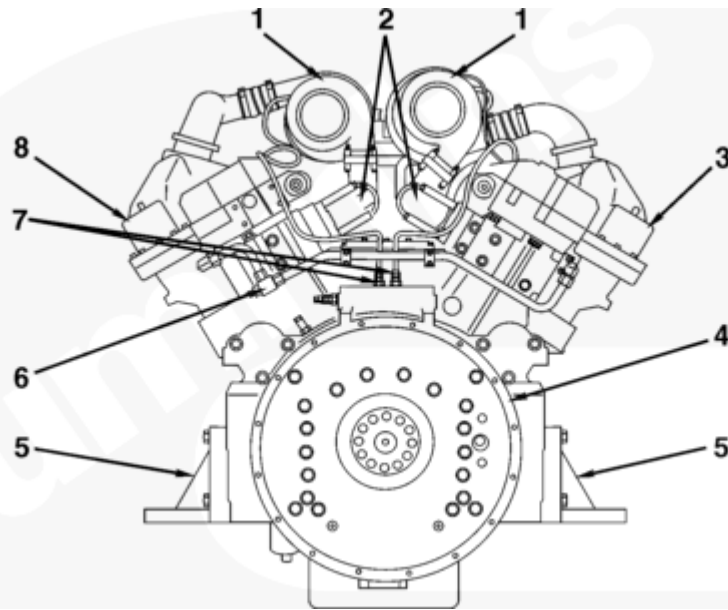
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QSK45 - Left Side View

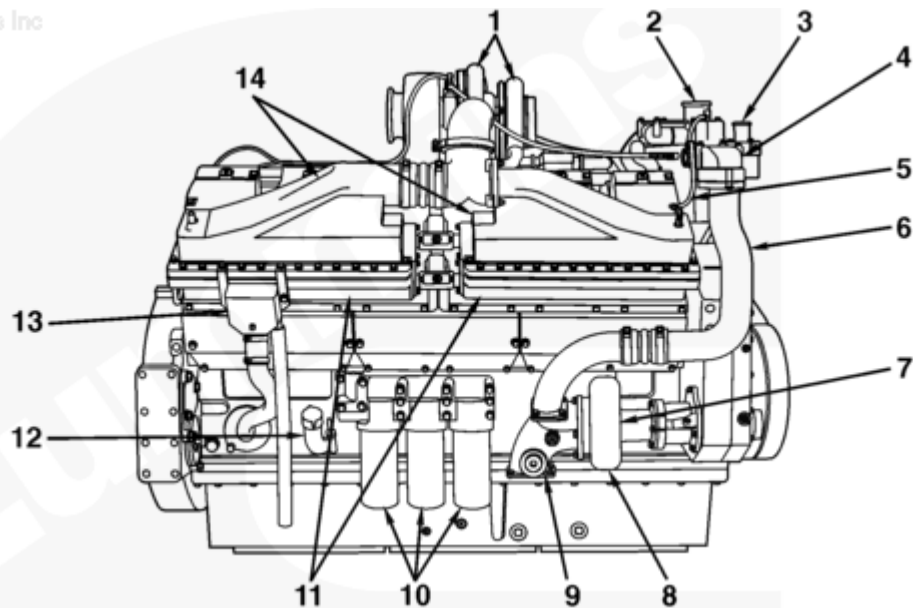
1. Turbocharger
2. Intake manifold covers
3. Aftercooler assemblies
4. Electronic control module (ECM)
5. Fuel inlet connection
6. Engine oil drains
7. Fuel filters
8. Crankcase breather
9. Fuel pump
10. LTA water pump
11. Thermostat housing support
12. Water filters
13. LTA water outlet connections.



00600017

QSK45 - Rear View

1. Turbochargers
2. Exhaust manifold
3. Right bank aftercooler assembly
4. Flywheel housing
5. Rear engine mount
6. Fuel return line
7. Turbocharger oil supply lines
8. Left bank aftercooler assembly.



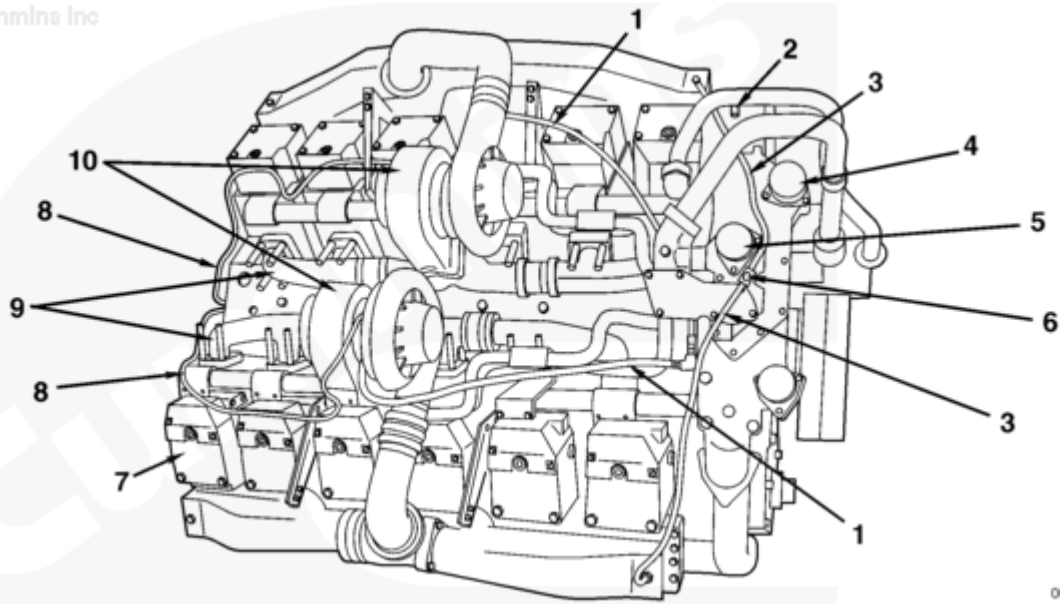
00600013

QSK45 - Right Side View

1. Turbochargers
2. LTA water outlet connections
3. Engine water outlet connections
4. Engine thermostat outlet

5. Aftercooler vent line
6. Water bypass tube
7. Water pump
8. Water pump drain
9. Coolant inlet connection
10. Oil filters
11. Aftercooler assemblies
12. Oil fill and dipstick tubes
13. Crankcase breather
14. Intake manifold cover.

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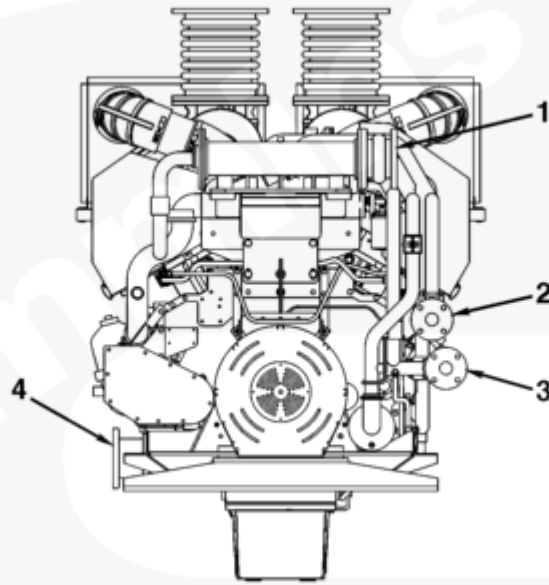


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QSK45 -Top View

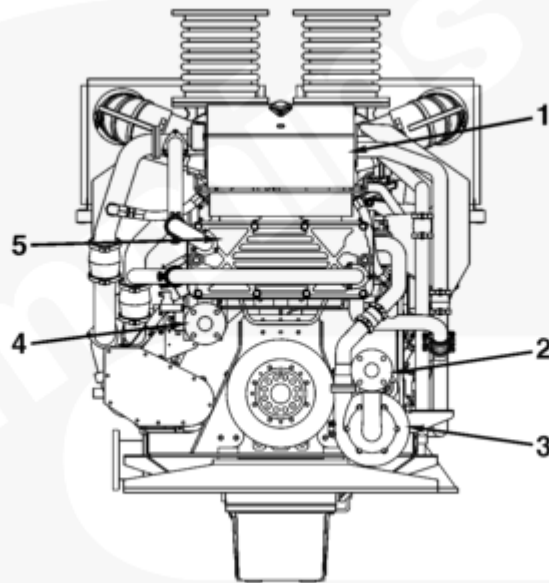
1. Coolant supply lines to turbocharger
2. Vent - test port
3. Aftercooler vent lines
4. Engine water outlet connections
5. LTA water outlet connection
6. Aftercooler vent line connection engine
7. Rocker lever cover
8. Oil supply lines to turbocharger
9. Exhaust manifolds
10. Turbochargers.

Marine Applications



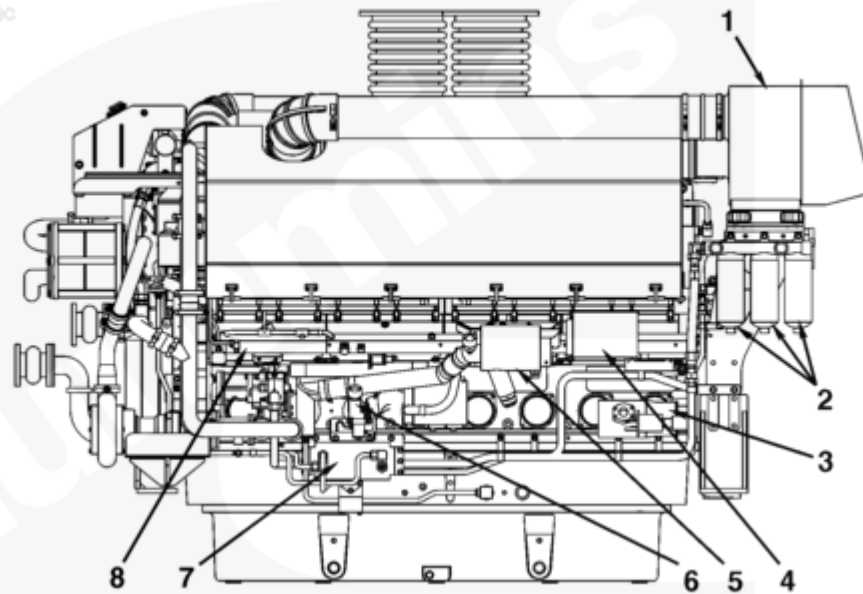
QSK60 Marine Keel Cooled - with Mechanically Actuated Injector - Front View

1. Engine coolant outlet connection to main engine cooler
2. LTA coolant outlet connection to LTA cooler
3. LTA coolant inlet connection from LTA cooler
4. Engine coolant inlet connection from main engine cooler.



QSK60 Marine Heat Exchanger Cooled - with Mechanically Actuated Injector - Front View

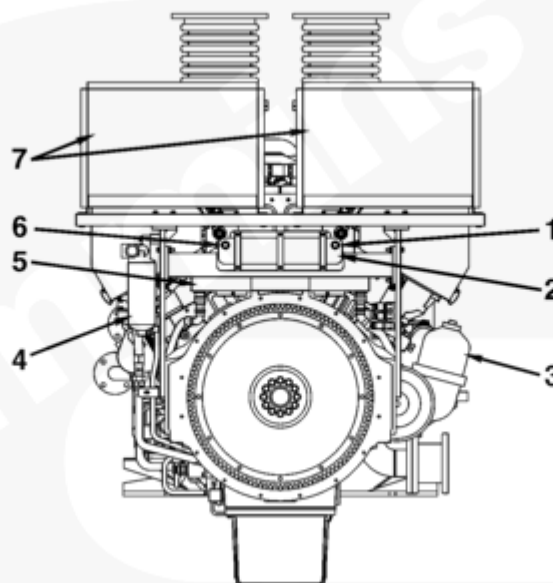
1. Expansion tank
2. Sea water inlet connection
3. Sea water pump
4. Sea water outlet connection
5. Heat exchanger.



00600216

QSK60 Marine - with Mechanically Actuated Injector - Left Side View

1. Air cleaner assembly
2. Fuel filters
3. Starter
4. Electronic control module (ECM)
5. Crankcase breather
6. Oil fill and dipstick tubes
7. Fuel cooler (optional)
8. Left bank wiring harness.



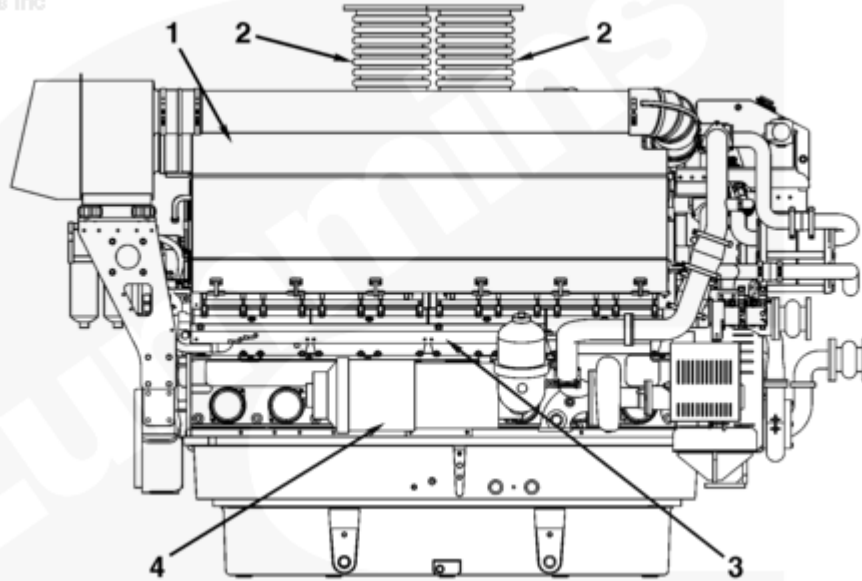
00600214

QSK60 Marine - with Mechanically Actuated Injector - Rear View

1. Gear oil inlet or outlet connection
2. Gear oil cooler (optional)
3. Eliminator™ lubricating oil filter assembly
4. Fuel filter

5. Rear engine harness
6. Gear oil inlet or outlet connection
7. Air filter assemblies.

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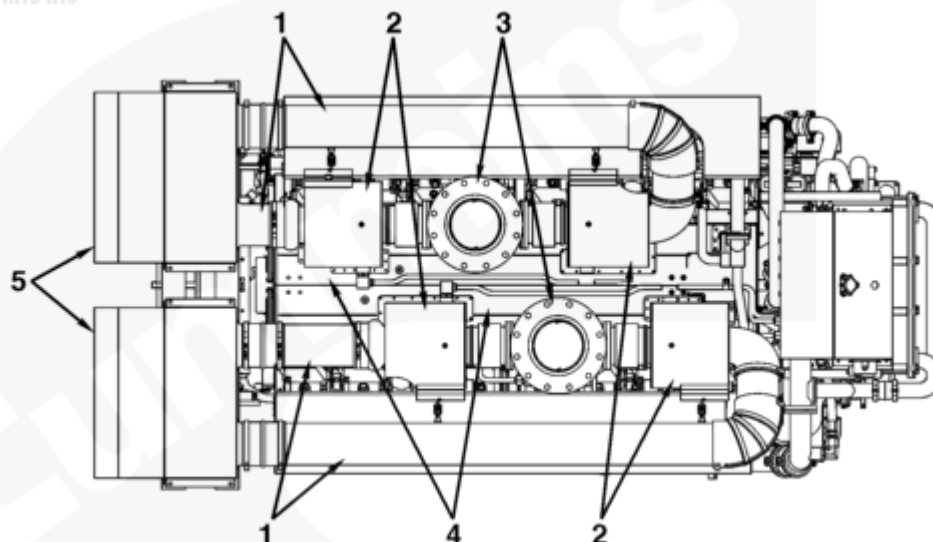


00600215

QSK60 Marine - with Mechanically Actuated Injector - Right Side View

1. Aftercooler heat shield
2. Exhaust outlet bellows
3. Right bank wiring harness
4. Eliminator™ lubricating oil filter assembly.

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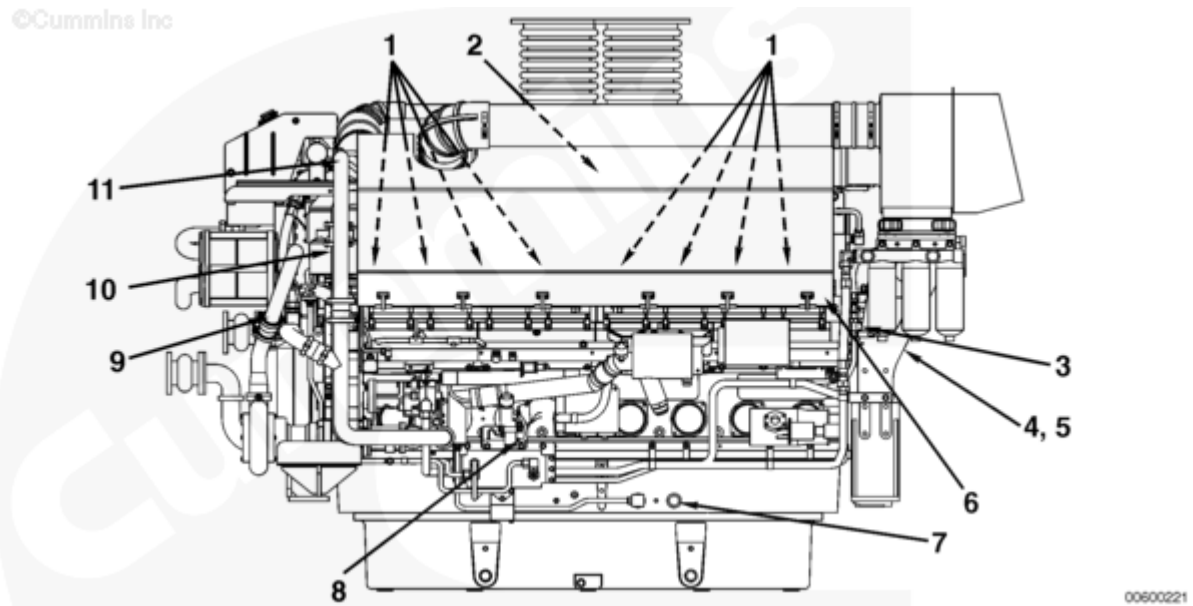


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QSK60 Marine - with Mechanically Actuated Injector - Top View

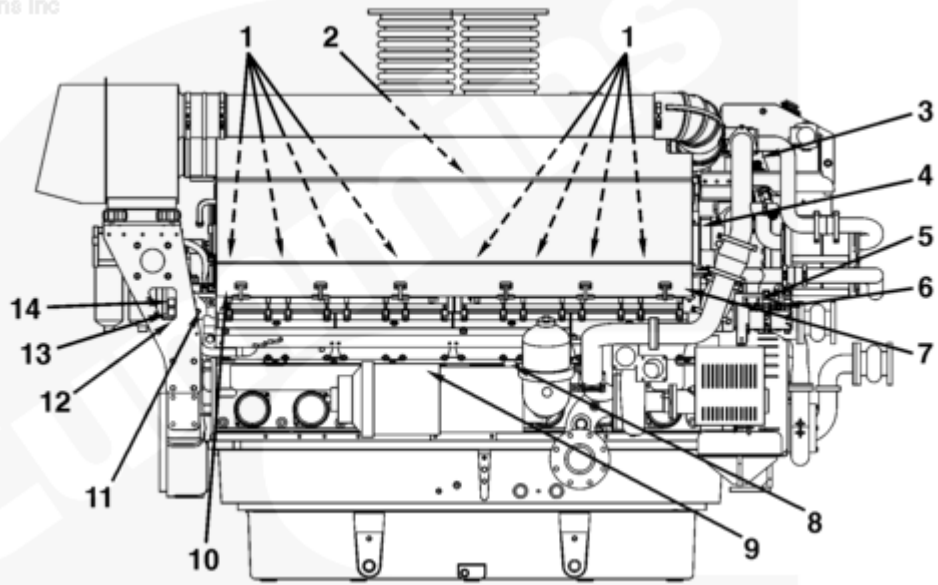
1. Air intake tubes
2. Turbocharger heat shields
3. Exhaust outlet connections
4. Exhaust manifold heat shields

5. Air cleaner assemblies.



QSK60 Marine Alarm, Safety, and Panel Sensors - with Mechanically Actuated Injector - Left Side View

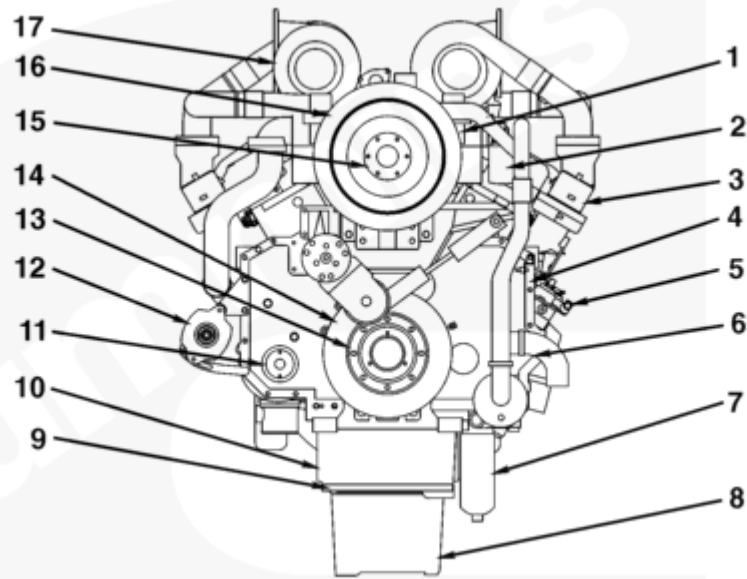
1. Cylinder exhaust gas temperature sensors
2. Exhaust stack temperature sensor
3. Engine speed sensor (safety system)
4. Fuel filter differential pressure sensor (located behind fuel filters)
5. Fuel filter outlet pressure sensor (located behind fuel filters)
6. Intake manifold temperature sensor
7. Oil temperature sensor
8. Fuel leakage sensor
9. Sea water pressure sensor (heat exchanger cooled engine **only**)
10. Coolant outlet temperature switch
11. Aftercooler water inlet temperature sensor.



QSK60 Marine Alarm, Safety, and Panel Sensors - with Mechanically Actuated Injector - Right Side View

1. Cylinder exhaust gas temperature sensors
2. Exhaust stack temperature sensor
3. Coolant level switch (heat exchanger engines **only**)
4. Coolant temperature sensor
5. Coolant inlet pressure switch
6. Coolant inlet pressure sensor
7. Intake manifold temperature sensor
8. Lubricating oil pressure sensor (before lubricating oil filter head)
9. Lubricating oil pressure sensor (after lubricating oil filter head)
10. Intake manifold temperature sensor
11. Engine speed sensor (for alarm system)
12. Lubricating oil pressure switch
13. Lubricating oil pressure switch, high speed range
14. Lubricating oil pressure switch, low speed range.

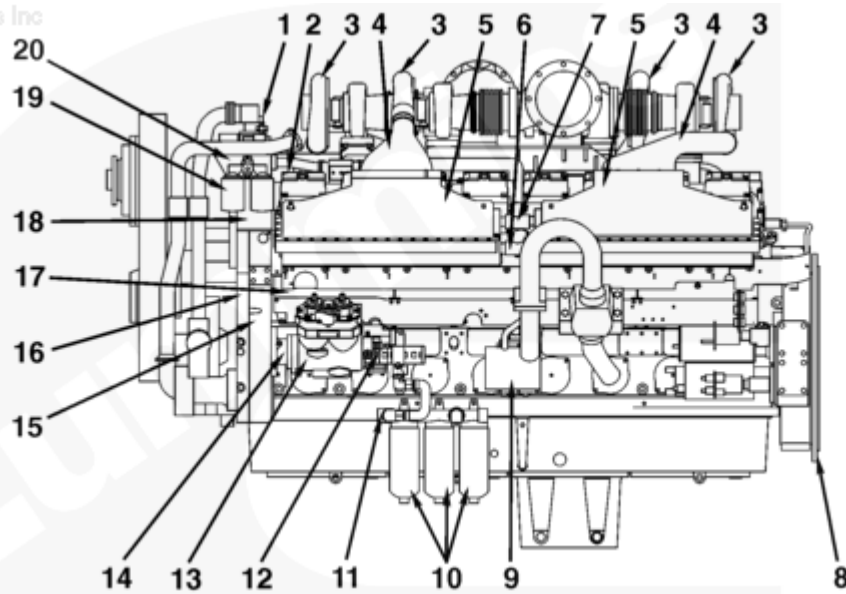
with Mechanically Actuated Injector



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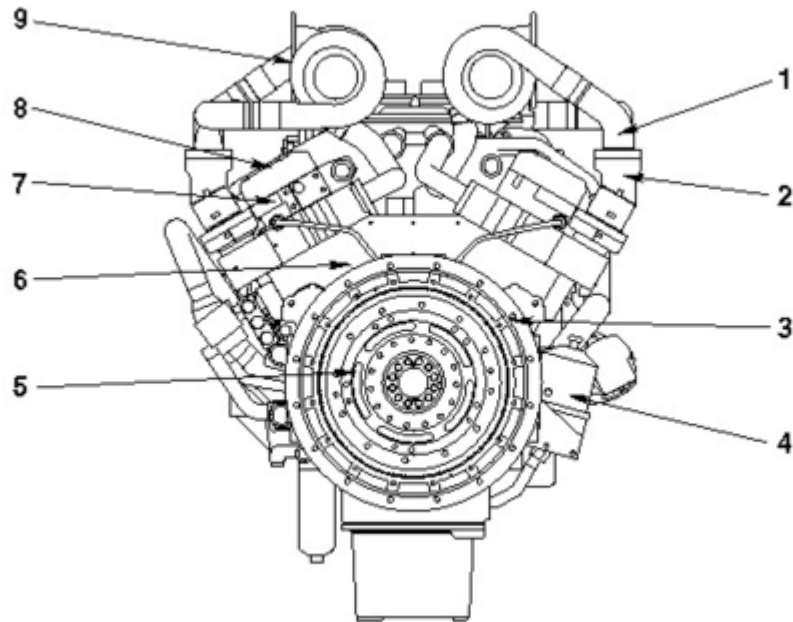
QSK60 - with Mechanically Actuated Injector - Front View

1. Thermostat housing
2. Coolant filter
3. Aftercooler assembly
4. Gear cover
5. Air compressor
6. Accessory drive pulley location (**not** shown)
7. Fuel filter
8. Oil pan sump
9. Oil pan adapter plate
10. Oil pan adapter
11. Alternator drive pulley
12. Alternator
13. Crankshaft pulley
14. Vibration dampers
15. Fan hub spacer
16. Fan hub pulley
17. Turbocharger.



QSK60 - with Mechanically Actuated Injectors - Left Side View

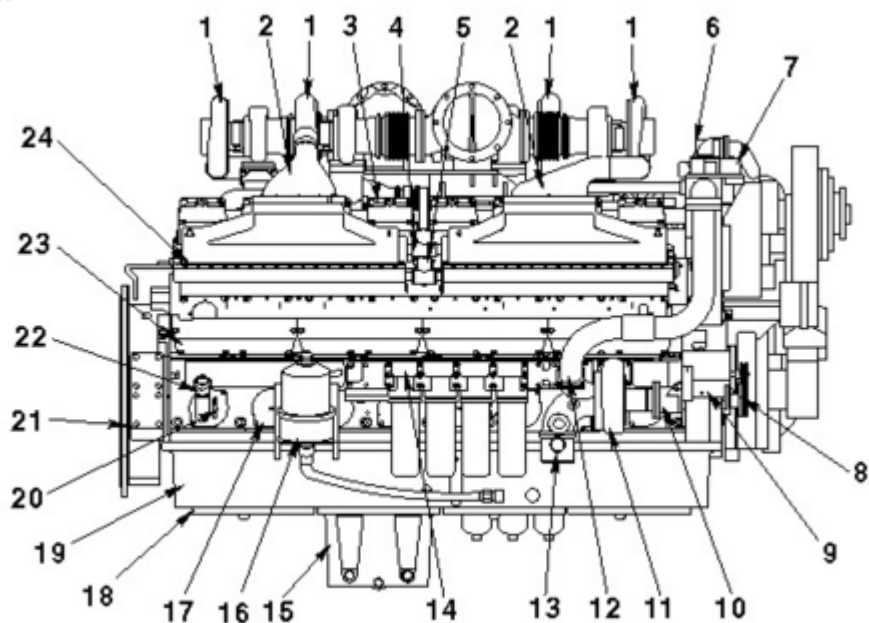
1. LTA water outlet connections
2. Rocker lever cover
3. Turbochargers
4. Air crossover connection
5. Aftercooler assemblies
6. Aftercooler water inlet tube
7. Aftercooler water outlet tube
8. Flywheel housing
9. Electronic control module (ECM)
10. Fuel filter
11. Fuel filter head
12. Fuel pump
13. Air compressor
14. Accessory drive
15. Front gear housing
16. Front gear cover
17. Cam follower cover
18. Thermostat housing support
19. Coolant filters
20. Thermostat housing.



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QS60 - with Mechanically Actuated Injectors - Rear View

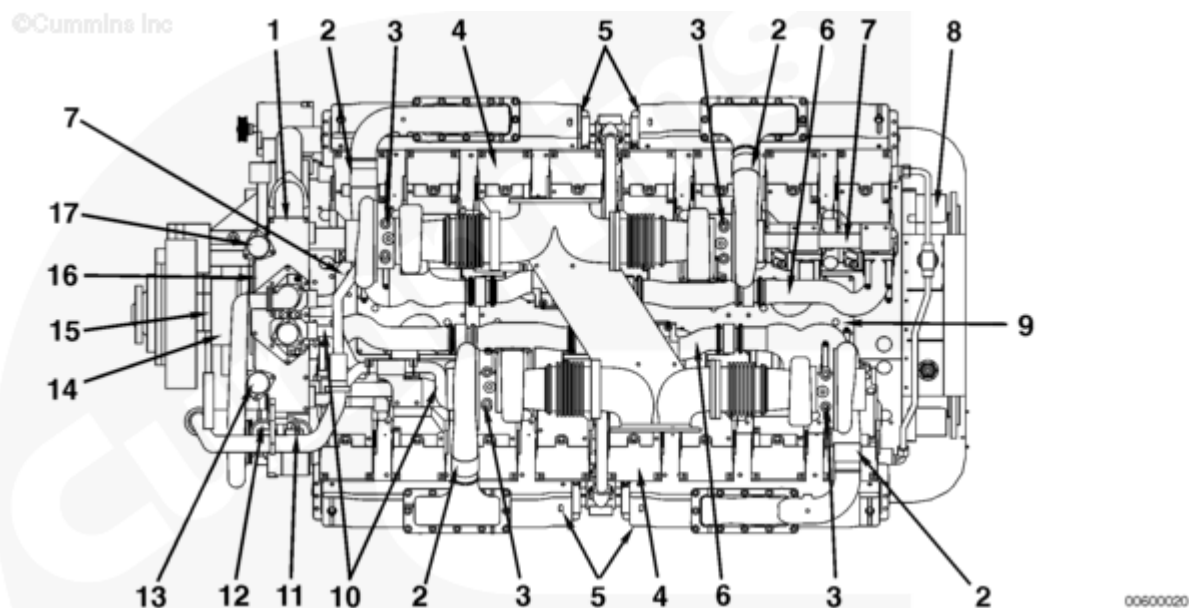
1. Air crossover connector
2. Aftercooler assembly
3. Flywheel housing
4. Centrifuge bypass filter
5. Flex plate
6. Cylinder block
7. Rocker lever housing
8. Rocker lever cover
9. Turbocharger.



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QS60 - with Mechanically Actuated Injectors - Right Side View

1. Turbochargers
2. Air crossover connections
3. Rocker lever cover
4. Aftercooler water outlet tube
5. Aftercooler water inlet tube
6. LTA water outlet connections
7. Engine water outlet connections
8. Alternator drive
9. Alternator
10. Water pump drive
11. Water pump
12. Water bypass tube
13. Water inlet connection
14. Oil filter head
15. Oil pan sump
16. Centrifuge bypass filter
17. Hand hole covers
18. Oil pan adapter plate
19. Oil pan adapter
20. Dipstick
21. Flywheel housing
22. Oil fill tube
23. Cam follower cover
24. Aftercooler assembly.

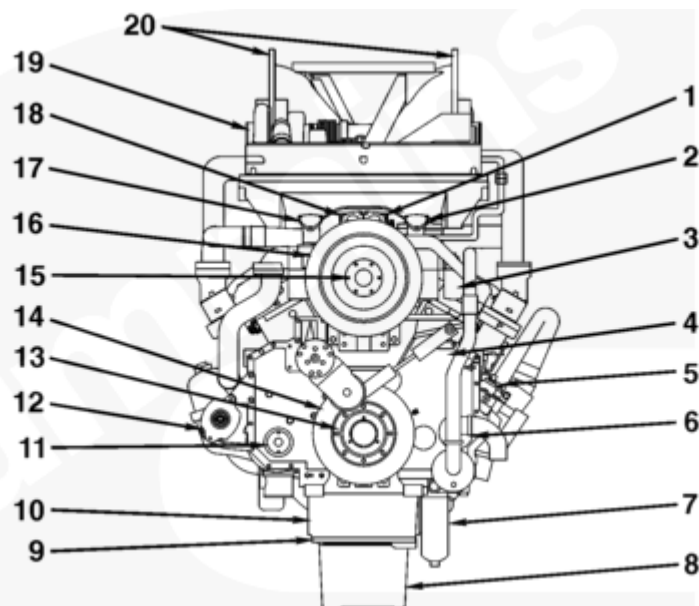


QSK60 - with Mechanically Actuated Injectors - Top View

1. Thermostat housing
2. Air crossover connections
3. Turbochargers
4. Rocker lever cover

5. Aftercooler assemblies
6. Exhaust manifolds
7. Water transfer tube
8. Flywheel housing
9. Oil cooler cover plate
10. Aftercooler water tube
11. Aftercooler water inlet temperature sensor
12. Coolant filter head
13. Left bank water outlet
14. Fan hub support
15. Fan hub shaft
16. Thermostat housing support
17. Right bank water outlet.

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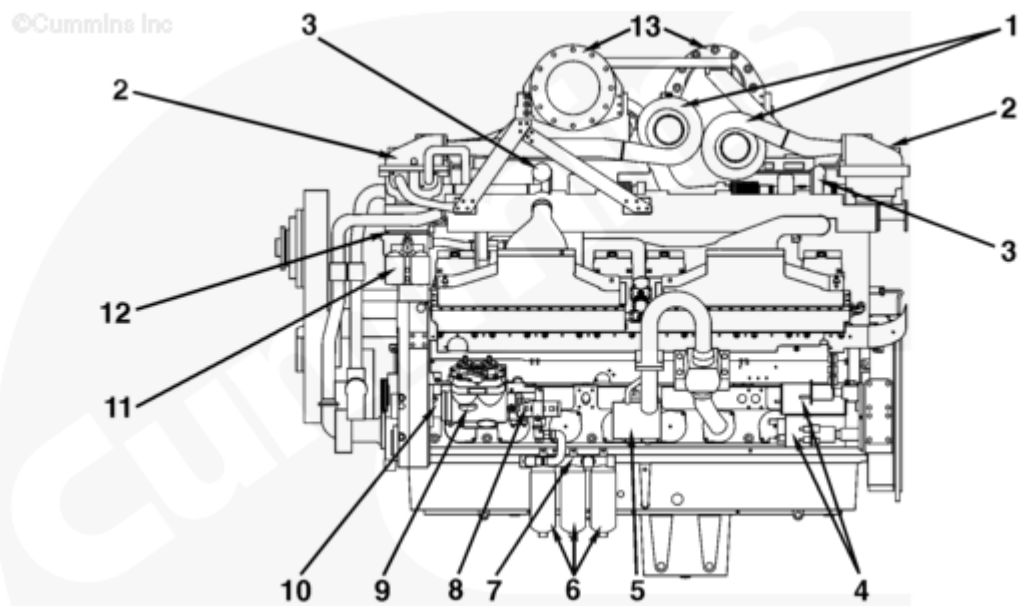


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QSK60, Two-Stage - with Mechanically Actuated Injectors - Front View

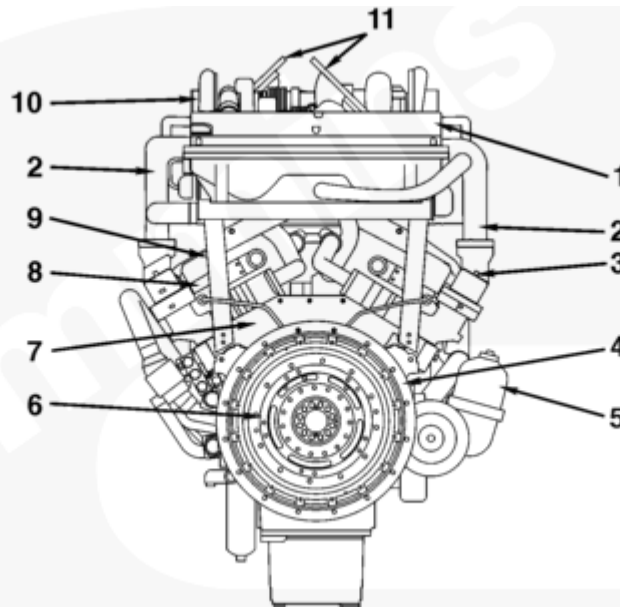
1. LTA water inlet connection
2. Left-Bank water outlet connection
3. Coolant filters
4. Gear cover
5. Air compressor
6. Accessory drive pulley
7. Fuel filters
8. Oil pan sump
9. Oil pan adapter plate
10. Oil pan adapter
11. Alternator drive pulley
12. Alternator
13. Crankshaft pulley
14. Vibration damper
15. Fan hub

16. Fan hub pulley
17. Right bank coolant outlet connection
18. LTA water outlet connection
19. Turbocharger
20. Exhaust outlet flanges.



QSK60, Two-Stage - with Mechanically Actuated Injectors - Left Side View

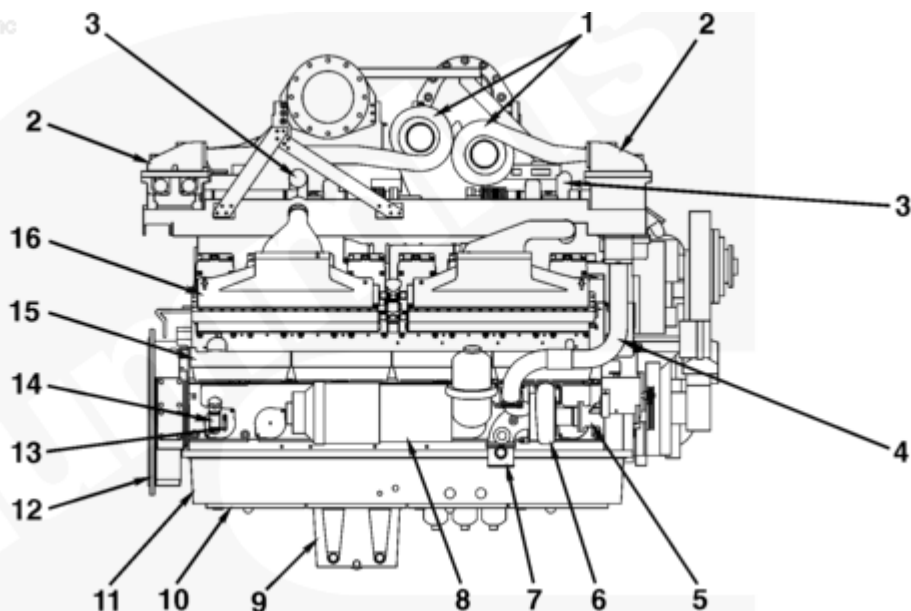
1. Low-Pressure turbochargers
2. Intercooler assemblies
3. High-Pressure turbochargers
4. Starters
5. Electronic control module (ECM)
6. Fuel filters
7. Fuel filter head
8. Fuel pump
9. Air compressor
10. Compressor drive
11. Coolant filters
12. Thermostat housing
13. Exhaust outlet flange.



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QSK60, Two-Stage - with Mechanically Actuated Injectors - Rear View

1. Intercooler assembly
2. Air crossover connection
3. Aftercooler assembly
4. Flywheel housing
5. Eliminator™ full-flow/bypass filtration system
6. Flex plate
7. Cylinder block
8. Rocker lever housing
9. Rocker lever cover
10. Turbocharger
11. Exhaust outlet flange.

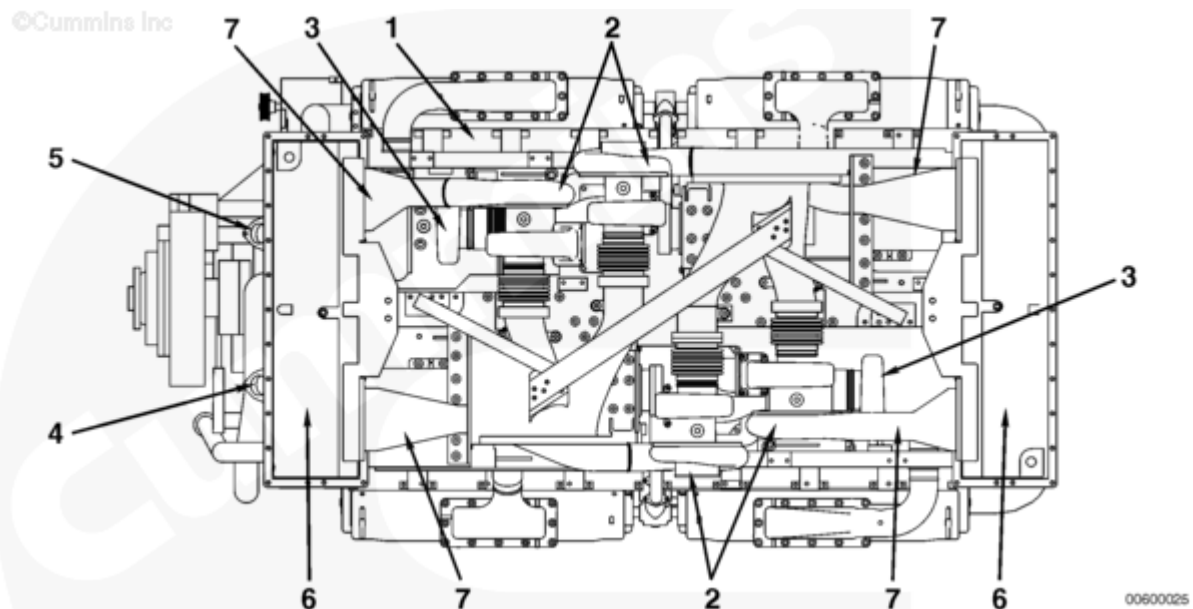


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QSK60, Two-Stage - with Mechanically Actuated Injectors - Right Side View

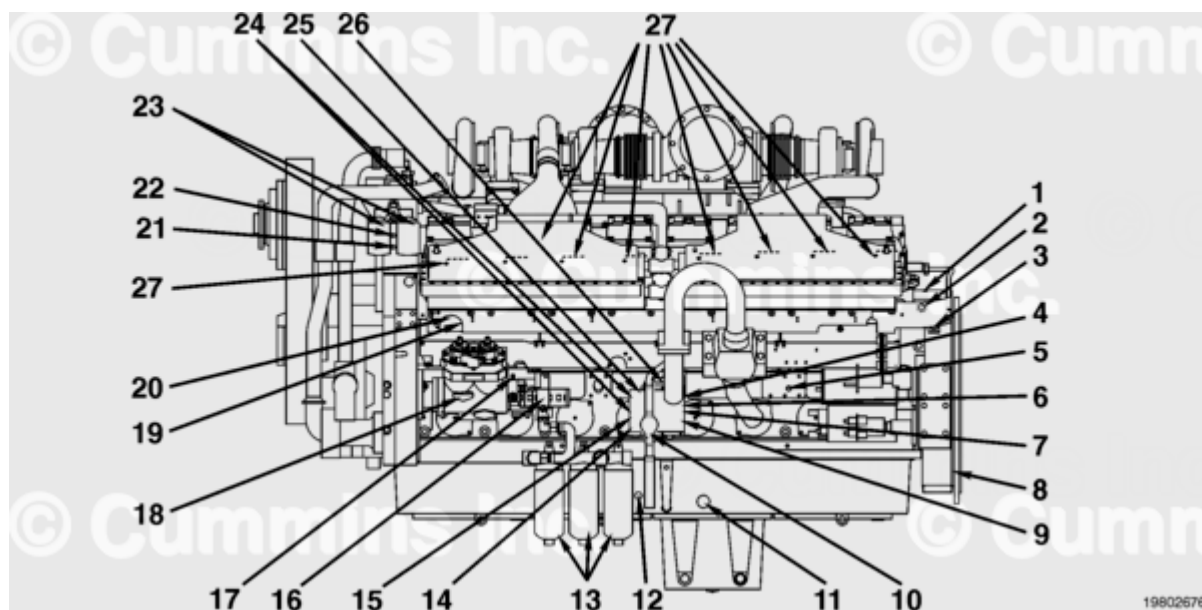
1. Low-Pressure turbochargers

2. Intercooler assemblies
3. High-Pressure turbochargers
4. Water bypass tube
5. Water pump drive
6. Water pump
7. Water pump inlet
8. Eliminator™ full-flow/bypass filtration system
9. Oil pan sump
10. Oil pan adapter plate
11. Oil pan adapter
12. Flywheel housing
13. Engine oil level dipstick
14. Oil fill tube
15. Cam follower cover
16. Aftercooler assembly.



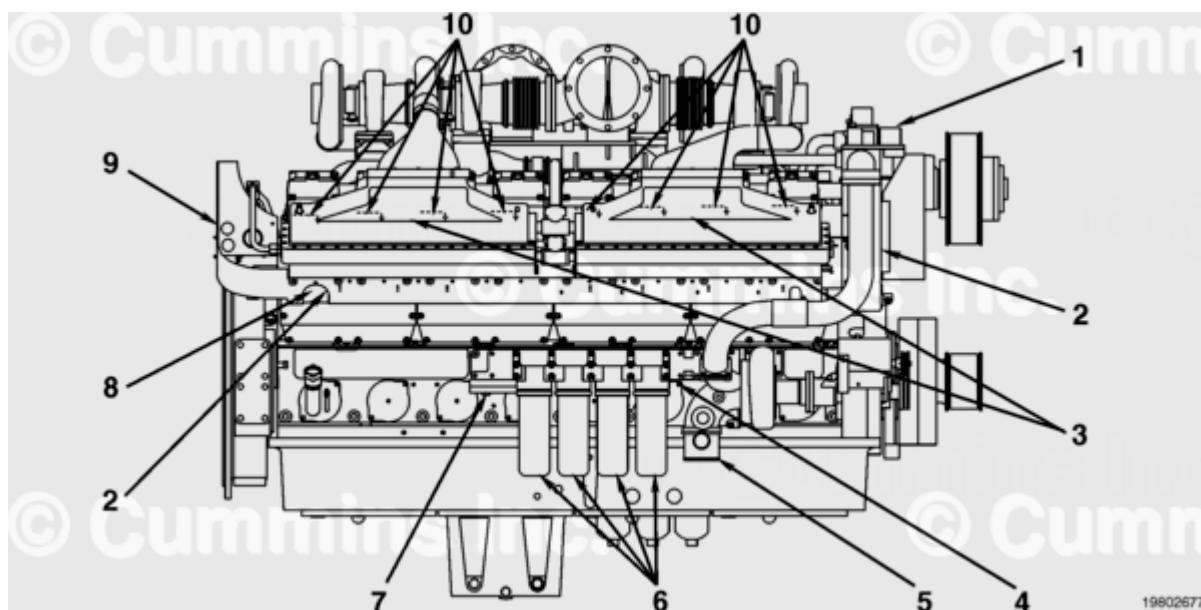
QSK60, Two-Stage - with Mechanically Actuated Injectors - Top View

1. Rocker lever cover
2. Low-Pressure turbochargers
3. High-Pressure turbochargers
4. Left bank water outlet
5. Right bank water outlet
6. Intercoolers
7. Air crossovers into intercoolers.



QSK60 (QSK45 Similar) - Left Bank Sensor Locations (Quantum™ and CENSE™) with Mechanically Actuated Injectors

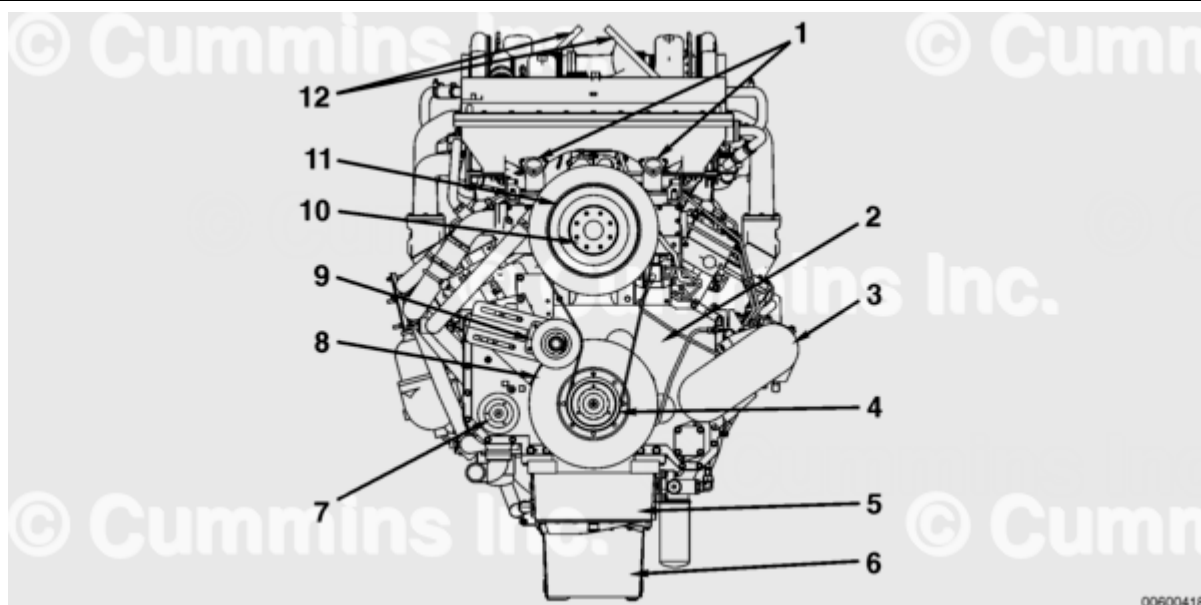
1. Compressor inlet temperature sensor (break out only - OEM installed)
2. Oil pressure sensor
3. Engine speed sensor
4. Timing pressure sensor
5. Service port
6. Ambient air pressure sensor
7. Fuel temperature sensor
8. Flywheel housing
9. Rail pressure sensor
10. Blowby sensor
11. Oil level sensor
12. Oil temperature sensor
13. Fuel filters
14. Rail actuator
15. Fuel shutoff valve
16. Fuel pump
17. Fuel pump quick-disconnect fitting
18. Air compressor
19. Intake manifold temperature sensor
20. Intake manifold pressure sensor
21. Coolant pressure sensor
22. Coolant temperature sensor
23. Coolant filters
24. Timing actuators
25. QUANTUM™ electronic control module (ECM)
26. Control valve body
27. Exhaust gas temperature sensors.



QSK60 (QSK45 Similar) - Right Bank Sensor Locations (Quantum™ and CENSE™) with Mechanically Actuated Injectors

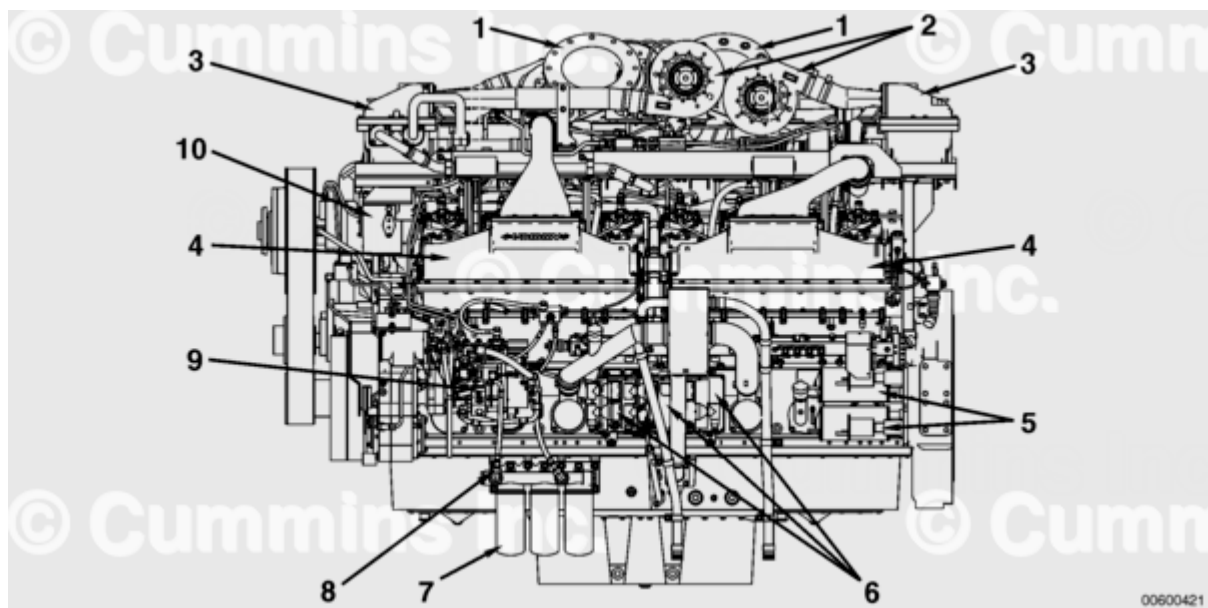
1. Coolant outlet
2. Intake manifold temperature sensor
3. Aftercooler assembly
4. Pre-filter oil pressure sensor
5. Coolant inlet
6. Oil filters
7. Post-filter pressure sensor
8. Intake manifold pressure sensor
9. CENSE™ electronic control module (ECM)
10. Exhaust gas temperature sensors.

with Electronically Actuated Injector



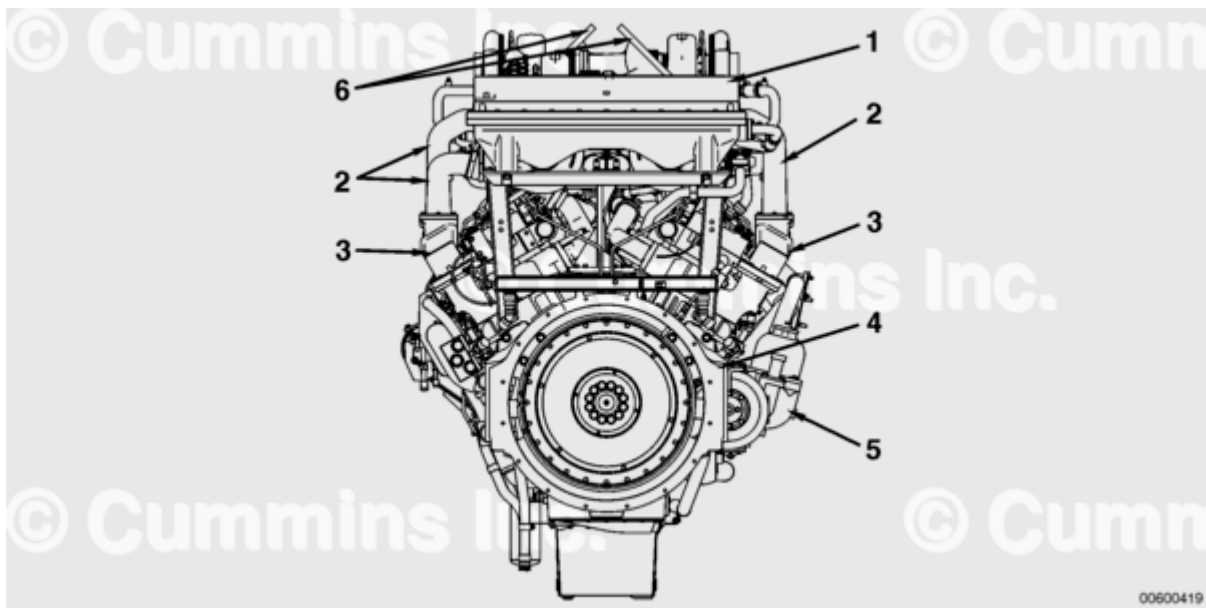
QSK60, Two Stage - with Electronically Actuated Injector - Front View

1. Coolant outlet connection
2. Gear cover
3. Accessory drive pulley cover
4. Crankshaft pulley
5. Oil pan adapter
6. Oil pan sump
7. Alternator drive pulley
8. Vibration damper
9. Fan belt tensioner
10. Fan hub
11. Fan hub pulley
12. Exhaust outlet flange.



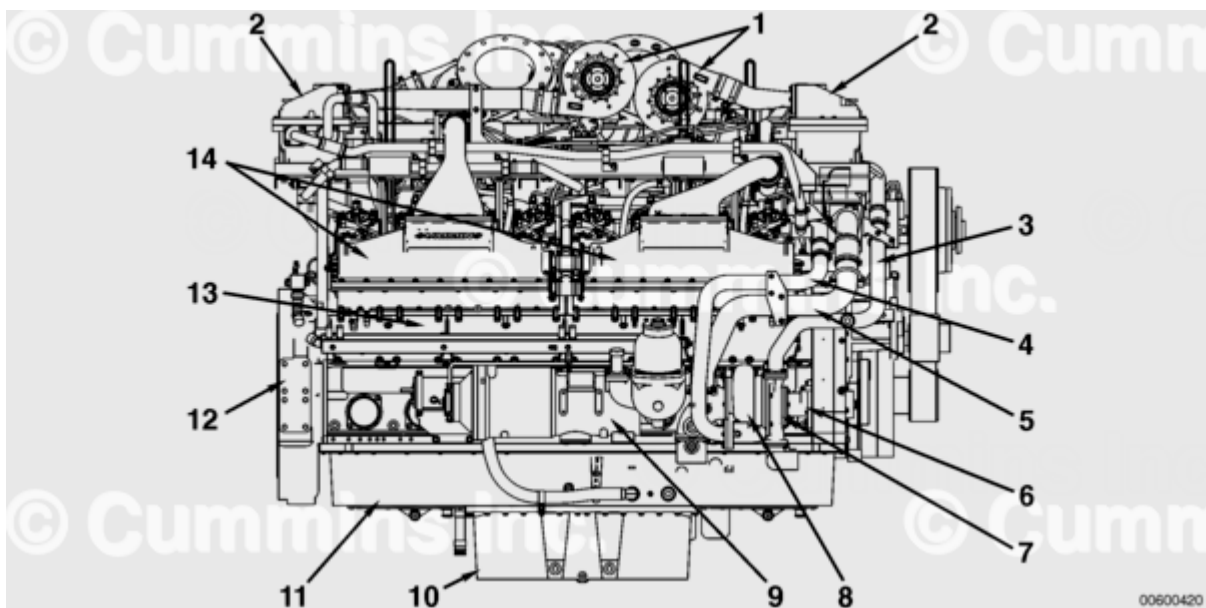
QSX60, Two Stage - with Electronically Actuated Injector - Left Side View

1. Exhaust outlet flange
2. Low pressure turbochargers
3. Intercooler assemblies
4. Aftercooler assemblies
5. Starters
6. Electronic control modules (ECMs)
7. Stage 2 fuel filters
8. Fuel filter head
9. Fuel pump
10. Thermostat housing.



QSK60, Two Stage - with Electronically Actuated Injector - Rear View

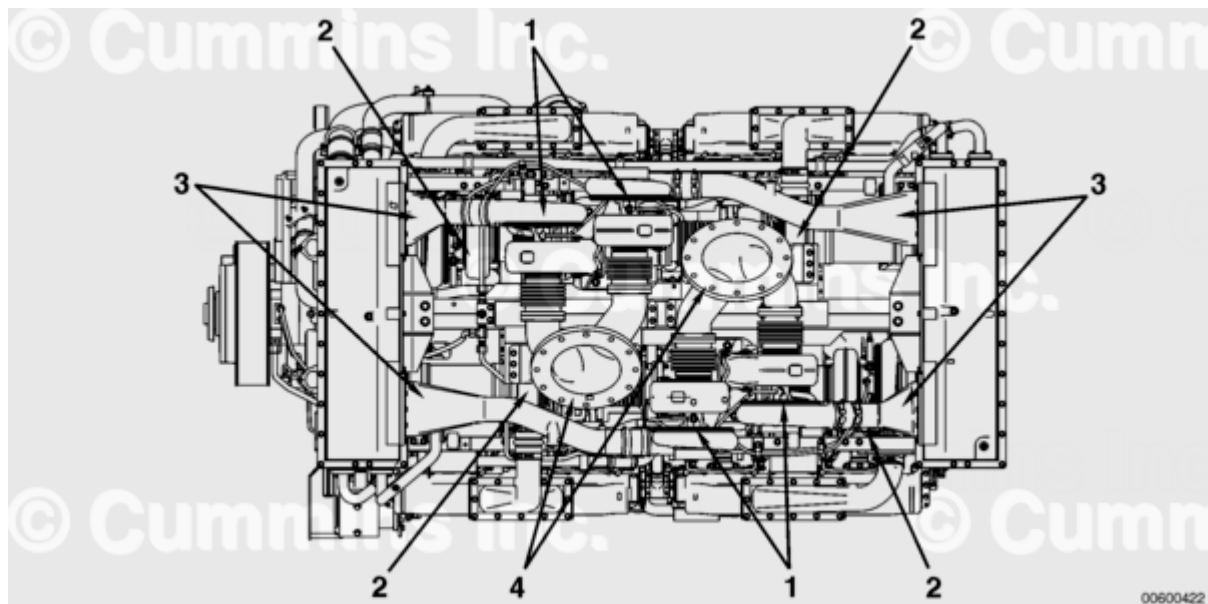
1. Rear intercooler assembly
2. Air crossover connections
3. Aftercooler assembly
4. Flywheel housing
5. Eliminator™ filtration system (optional)
6. Exhaust outlet flange.



QSK60, Two Stage - with Electronically Actuated Injector - Right Side View

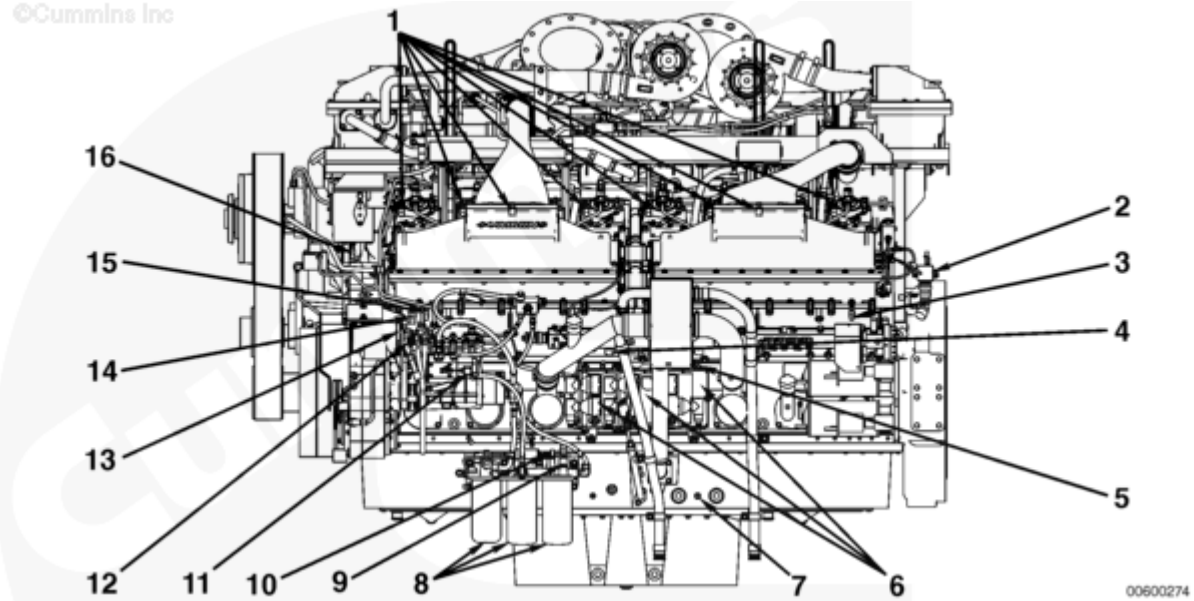
1. Low pressure turbochargers
2. Intercooler assemblies
3. Low temperature aftercooler bypass
4. Low temperature aftercooler outlet from water pump
5. Engine coolant bypass
6. Water pump drive

7. Low temperature aftercooler water pump
8. Jacket water pump
9. Eliminator™ filtration system (optional)
10. Oil pan sump
11. Oil pan adapter
12. Flywheel housing
13. Cam follower cover
14. Aftercooler assemblies.



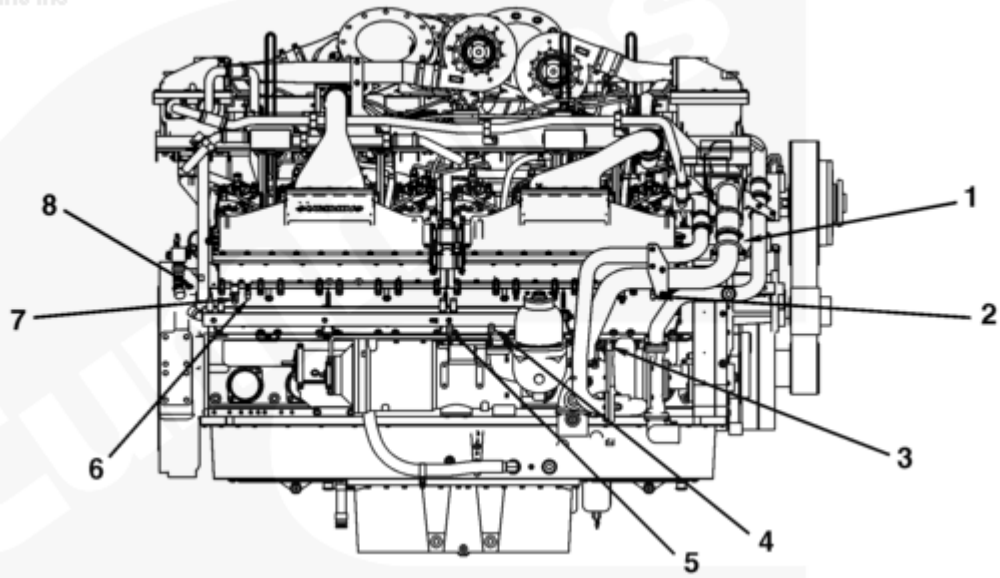
QSK60, Two Stage - with Electronically Actuated Injector - Top View

1. Low pressure turbochargers (4)
2. High pressure turbochargers (4)
3. Air crossover into intercooler
4. Exhaust outlet flange.



QSK60, Two Stage - with Electronically Actuated Injector - Left Bank Sensor Locations

1. Exhaust gas temperature sensors (eight on left bank)
2. Engine oil rifle pressure 1 sensor
3. Intake manifold 3 temperature sensor
4. Service port
5. Barometric pressure sensor
6. Electronic control modules (ECMs)
7. Engine oil temperature sensor 1
8. Water in fuel indicator sensors
9. Fuel delivery pressure sensor
10. Engine fuel temperature sensor 1
11. Fuel pump pressurizing assembly 1
12. Engine camshaft speed/position sensor
13. Injector metering rail 1 pressure sensor
14. Intake manifold 1 pressure sensor
15. Intake manifold 1 temperature sensor
16. Engine coolant temperature 1 sensor.

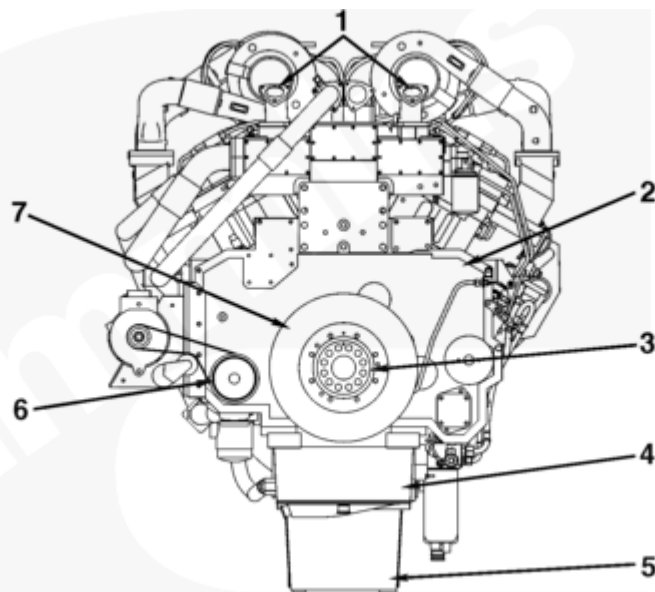


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QSK60, Two Stage - with Electronically Actuated Injector - Right Bank Sensor Locations

1. Crankcase pressure sensor
2. Intake manifold 2 temperature sensor
3. Coolant pressure sensor
4. Pre-oil filter pressure sensor
5. Post-oil filter pressure sensor
6. Intake manifold 4 temperature sensor
7. Intake manifold 2 pressure sensor
8. Engine crankshaft speed/position sensor.

* Exhaust gas temperature sensors (eight on right bank, **not** shown)



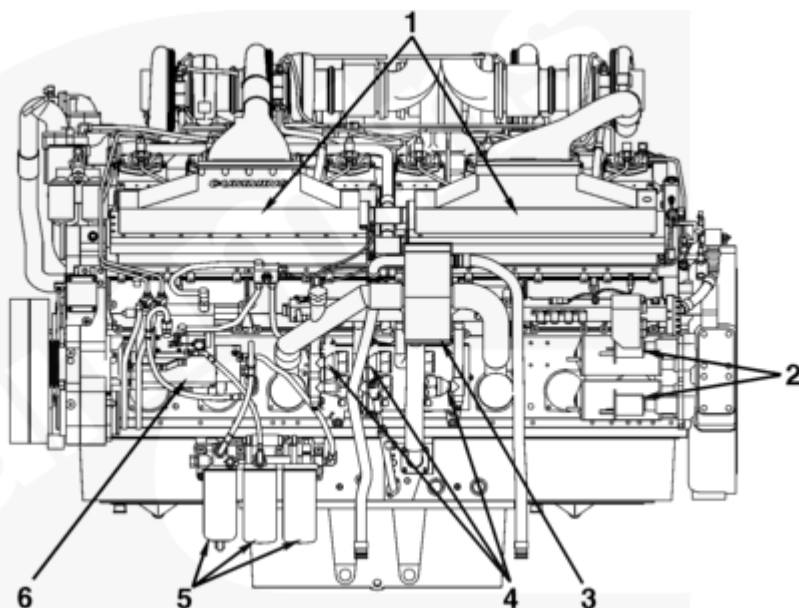
00600362

QSK60, Single Stage - with Electronically Actuated Injector - Front View

1. Coolant outlet connection
2. Gear cover
3. Crankshaft pulley

4. Oil pan adapter
5. Oil pan sump
6. Alternator drive pulley
7. Vibration damper.

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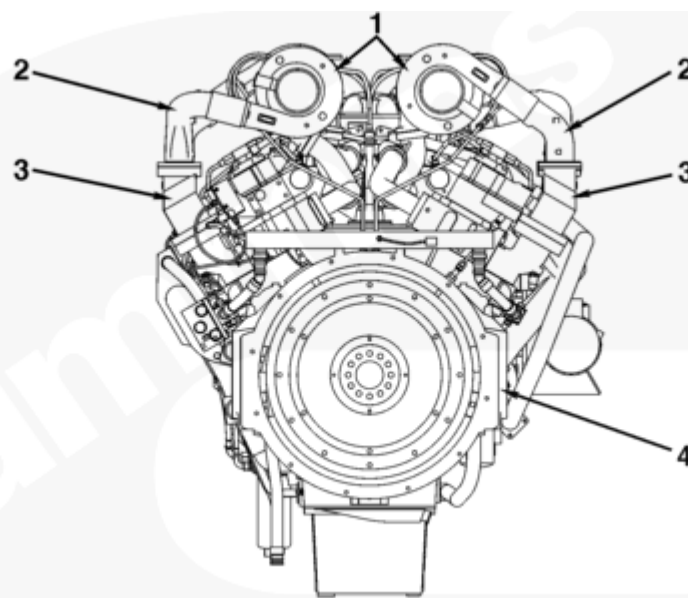


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QSK60, Single Stage - with Electronically Actuated Injector - Left Side View

1. Aftercooler assemblies
2. Starters
3. Breather
4. Electronic control modules (ECMs)
5. Stage two fuel filters
6. Fuel pump.

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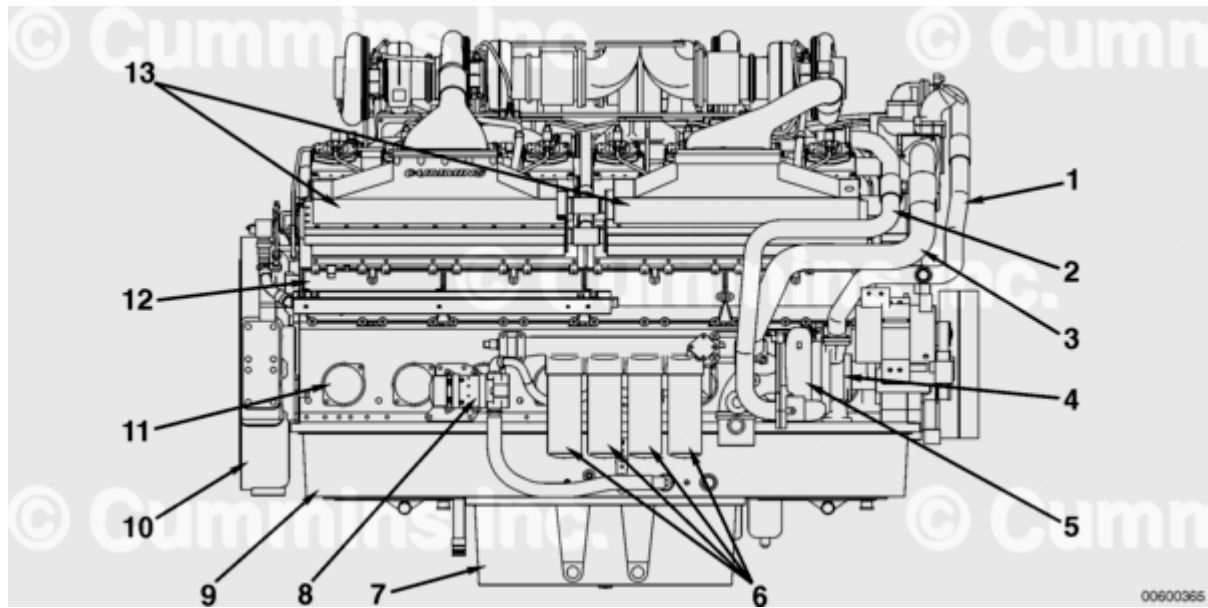


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QSK60, Single Stage - with Electronically Actuated Injector - Rear View

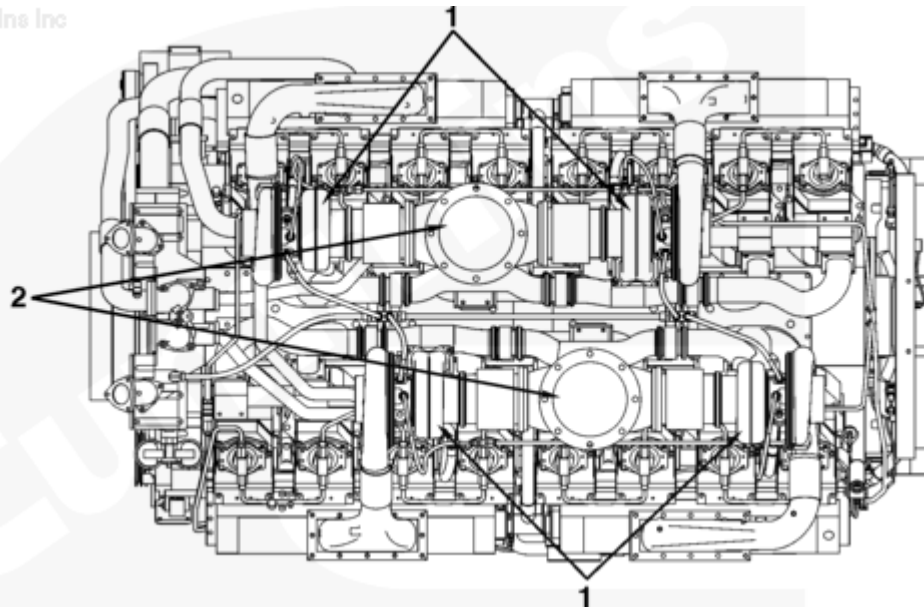
1. Rear turbochargers

2. Air crossover connections
3. Aftercooler assembly
4. Flywheel housing.



QSK60, Single Stage - with Electronically Actuated Injector - Right Side View

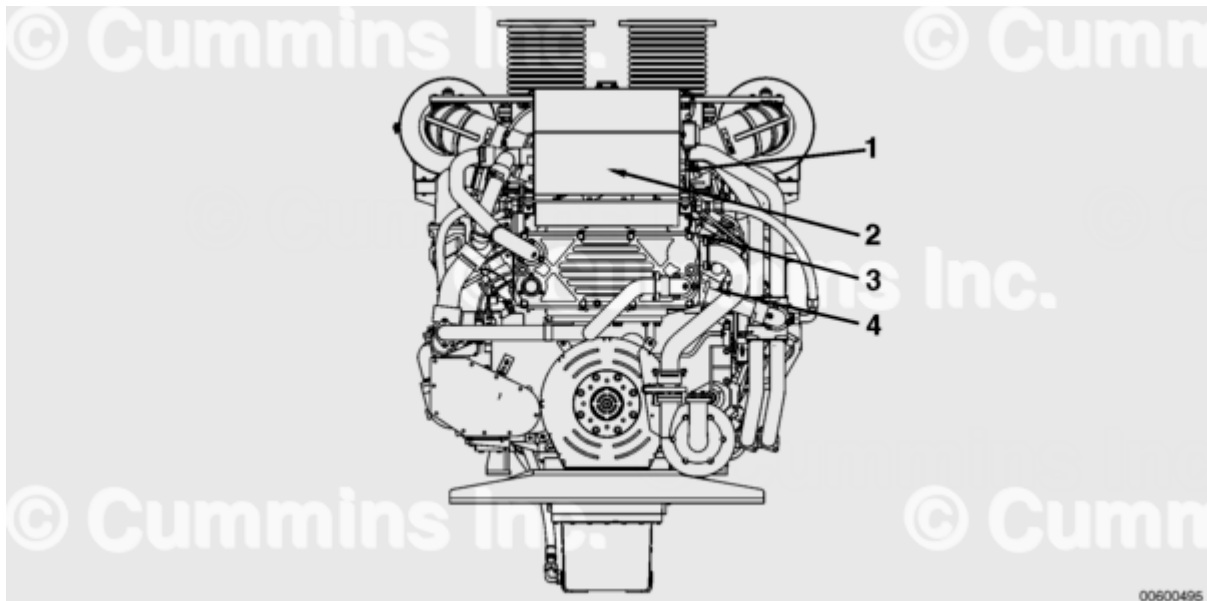
1. Low temperature aftercooler bypass
2. Low temperature aftercooler outlet from water pump
3. Engine coolant bypass
4. Low temperature aftercooler water pump
5. Jacket water coolant pump
6. Lubricating oil filters (4)
7. Oil pan sump
8. Pre-lubricating pump
9. Oil pan adapter
10. Flywheel housing
11. Hand hole cover
12. Cam follower cover
13. Aftercooler assemblies.



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QSK60, Single Stage - with Electronically Actuated Injector - Top View

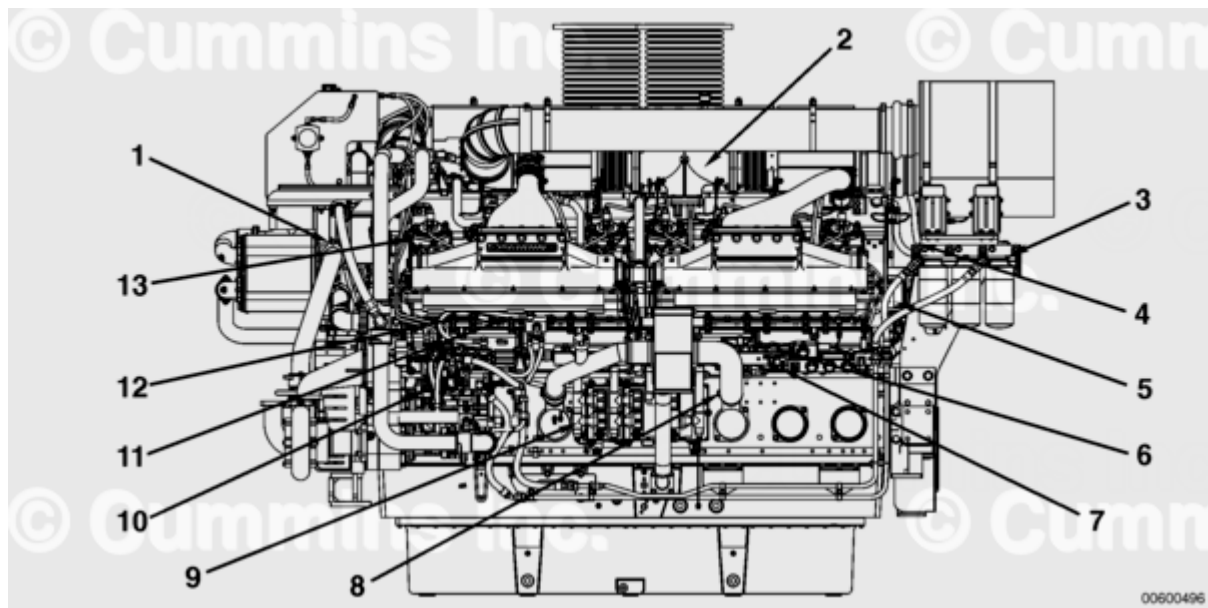
1. Turbochargers (4)
2. Exhaust outlet flanges (2).



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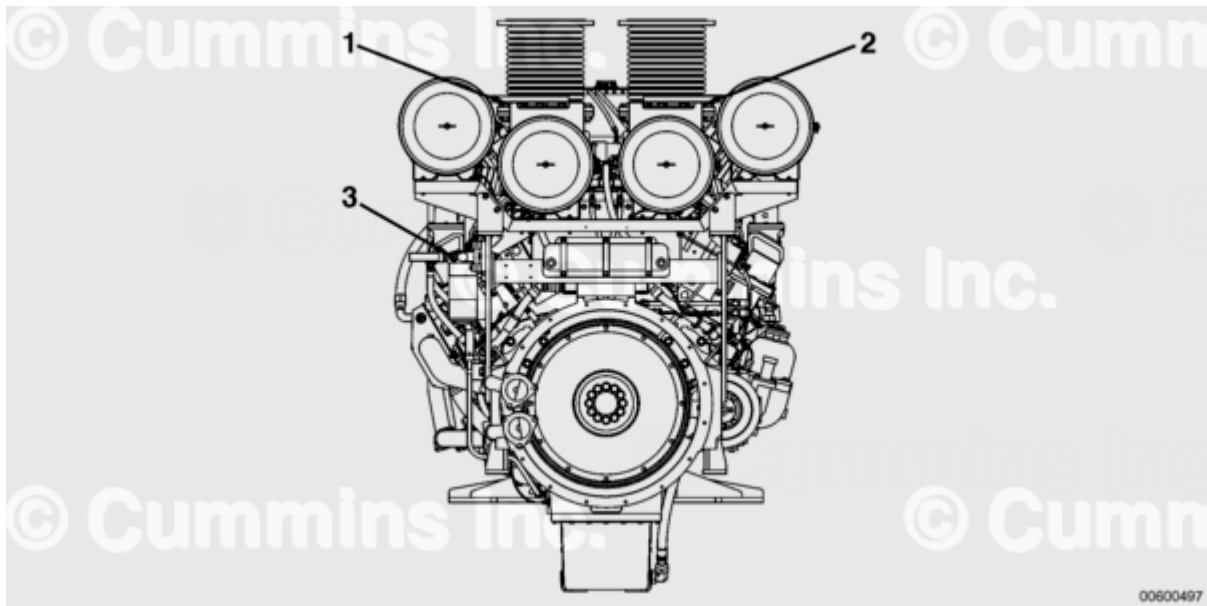
QSK60 Marine, Heat Exchanger Cooled - with Electronically Actuated Injector - Front View

1. Engine coolant level sensors
2. Coolant sight glass
3. Engine coolant temperature sensor
4. Camshaft position sensor.



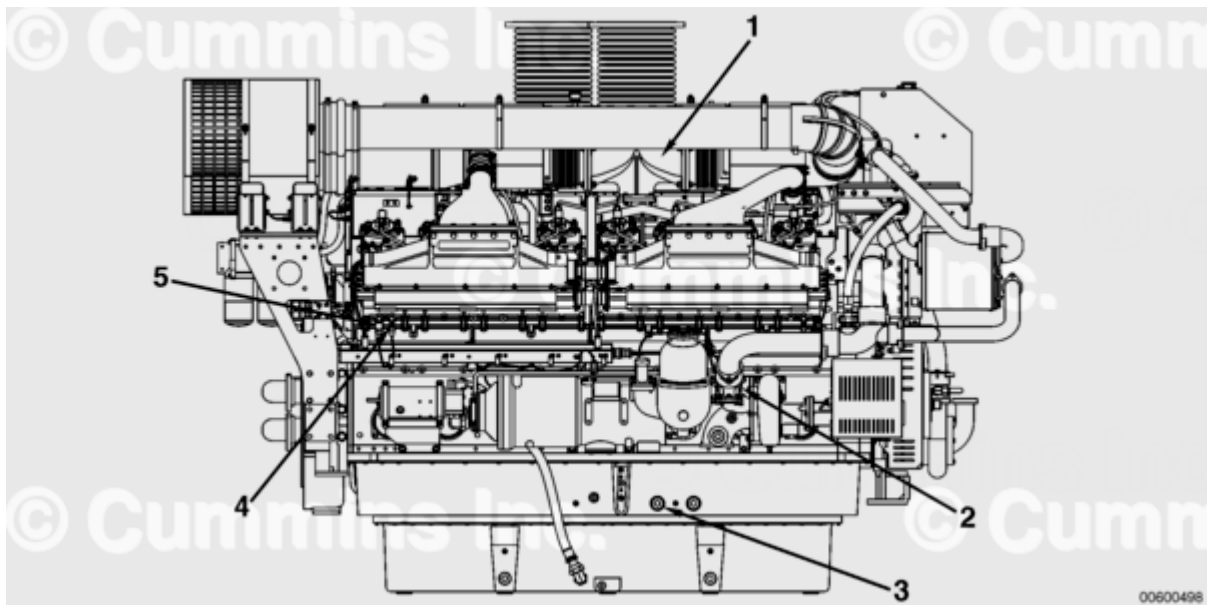
QSK60 Marine, Heat Exchanger Cooled - with Electronically Actuated Injector - Left View

1. Coolant temperature sensor
2. Engine exhaust gas stack temperature sensor, left manifold
3. Fuel supply pressure sensor
4. Fuel temperature sensor
5. Engine speed sensor (behind flywheel housing)
6. Ambient air pressure sensor
7. Oil rifle pressure sensor
8. Pre-oil filter pressure sensor
9. Electronic control modules (ECMs) (see last page of Engine Diagrams for identification)
10. Fuel accumulator pressure sensor
11. Intake manifold temperature sensor, left bank front
12. Intake manifold pressure sensor (left bank front)
13. Exhaust gas temperature sensors (one per cylinder, located in the exhaust port outlet of the cylinder head).



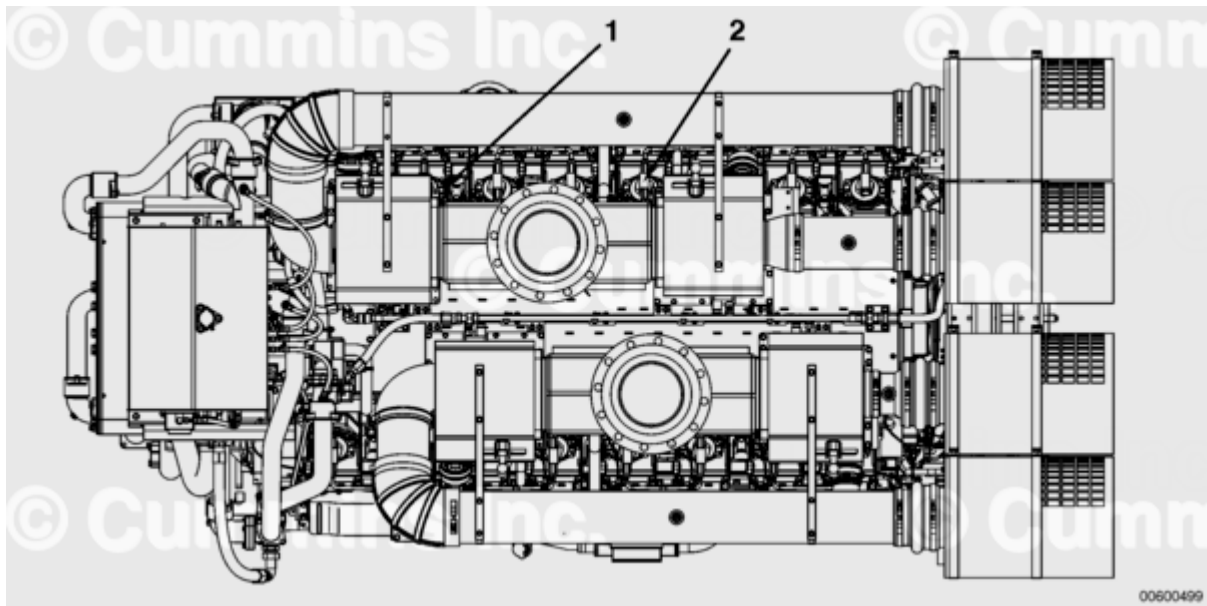
QSK60 Marine, Heat Exchanger Cooled - with Electronically Actuated Injector - Rear View

1. Engine exhaust gas stack temperature sensor, left manifold
2. Engine exhaust gas stack temperature sensor, right manifold
3. Fuel supply pressure sensor.



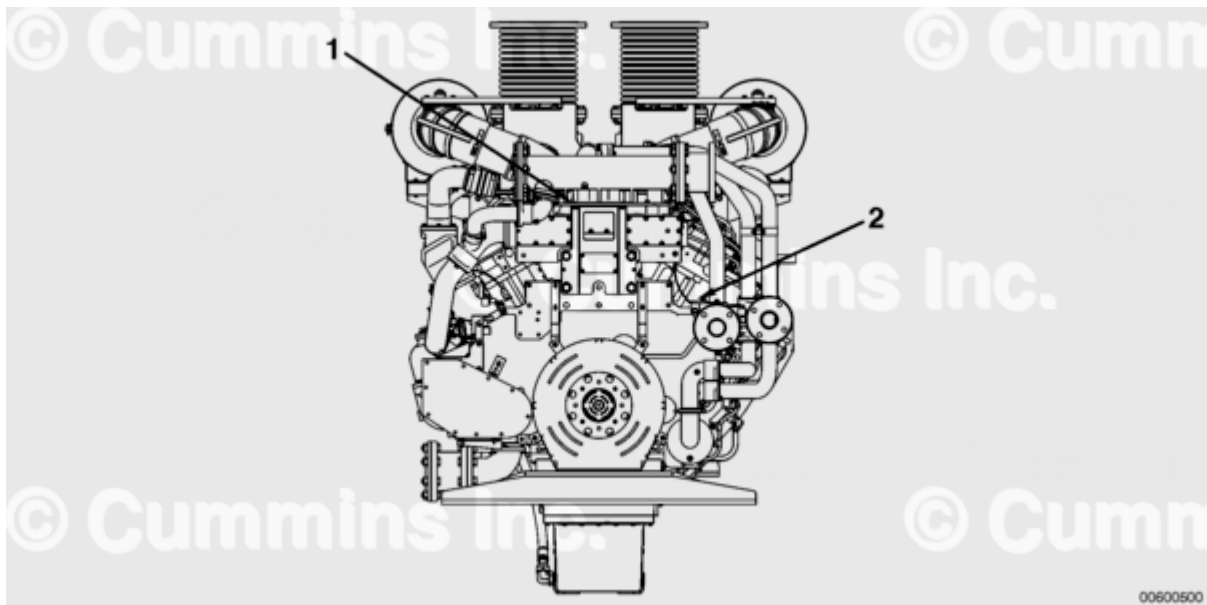
QSK60 Marine, Heat Exchanger Cooled (Aftercooler Shields Removed) - with Electronically Actuated Injector - Right View

1. Engine exhaust gas stack temperature sensor, right manifold
2. Engine coolant pressure sensor
3. Lubricating oil temperature sensor
4. Intake manifold pressure sensor (rear right bank)
5. Intake manifold temperature sensor (rear right bank).



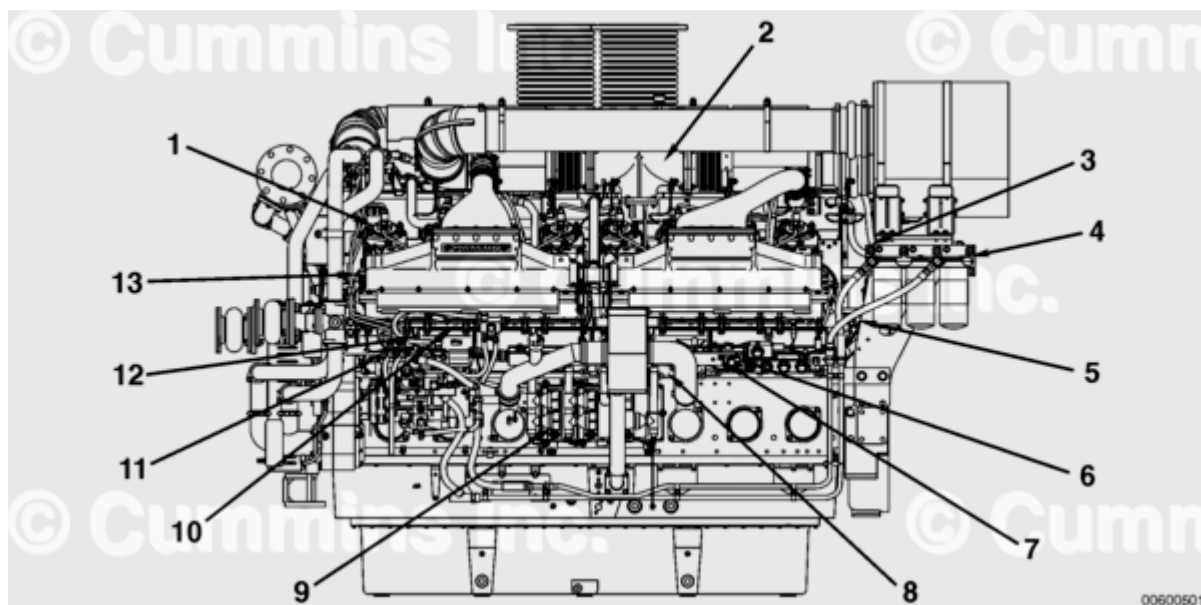
QSK60 Marine, Heat Exchanger Cooled - with Electronically Actuated Injector - Top View

1. Exhaust gas temperature sensor (one per cylinder, located in the exhaust port outlet of the cylinder head)
2. Injector solenoid (one per cylinder, located on top of the rocker housing cover).



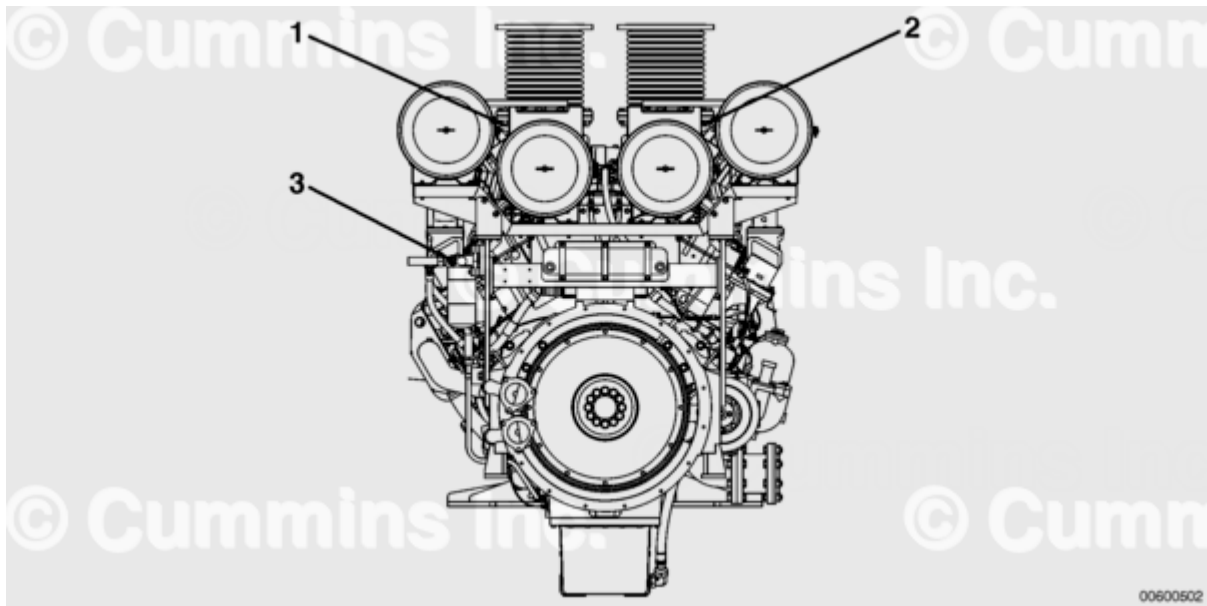
QSK60 Marine Keel Cooled - with Electronically Actuated Injector - Front View

1. Engine coolant temperature sensor
2. Camshaft position sensor.



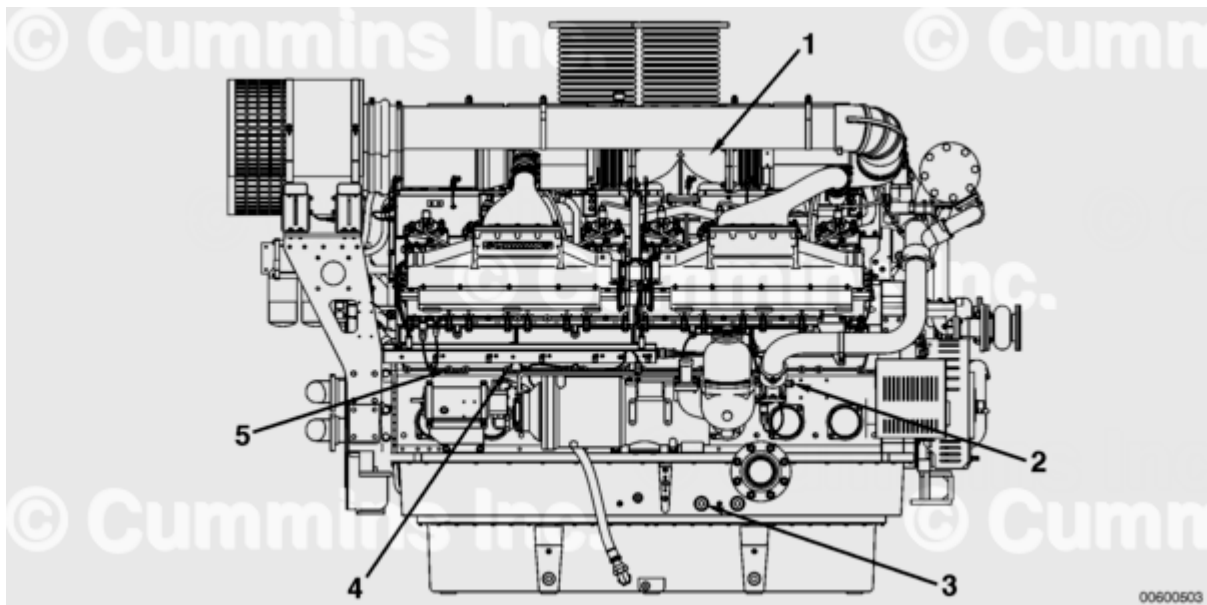
QSK60 Marine Keel Cooled - with Electronically Actuated Injector - Left View

1. Exhaust gas temperature sensors (one per cylinder, located in the exhaust port outlet of the cylinder head)
2. Engine exhaust gas stack temperature sensor, left manifold
3. Fuel temperature sensor
4. Fuel supply pressure sensor
5. Engine speed sensor (behind flywheel housing)
6. Oil rifle pressure sensor
7. Pre-oil filter pressure sensor
8. Ambient air pressure sensor
9. Electronic control modules (ECMs) (see last page of engine diagrams for identification)
10. Intake manifold temperature sensor, left bank front
11. Fuel accumulator pressure sensor
12. Intake manifold pressure sensor (left front)
13. Coolant temperature sensor.



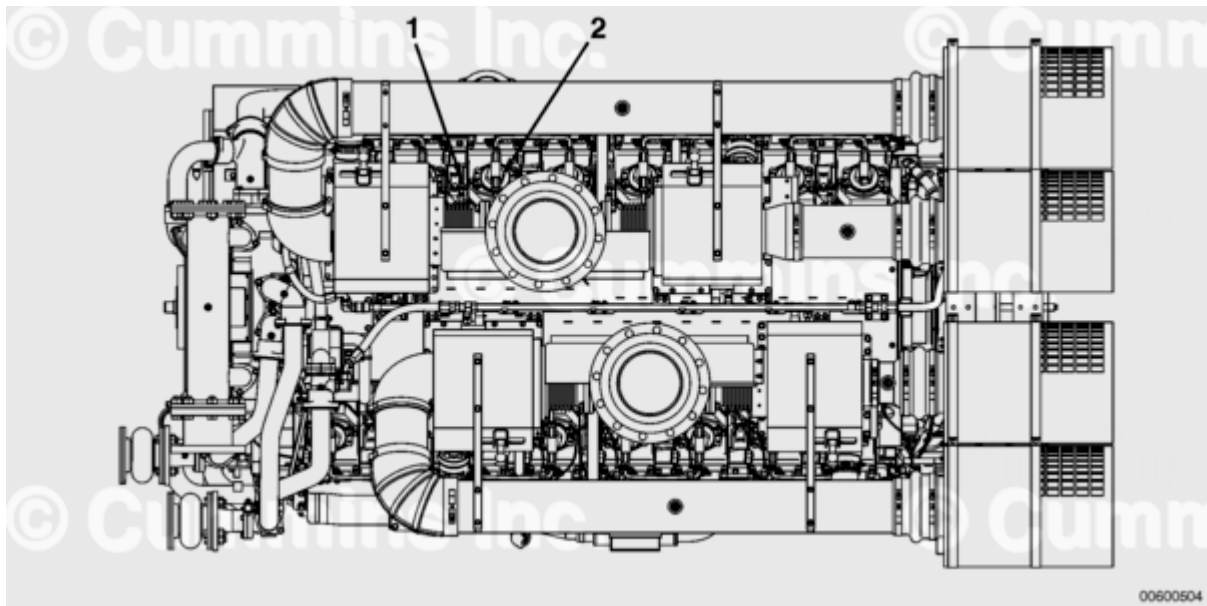
QSK60 Marine Keel Cooled - with Electronically Actuated Injector - Rear View

1. Engine exhaust gas stack temperature sensor, left manifold
2. Engine exhaust gas stack temperature sensor, right manifold
3. Fuel supply pressure sensor.



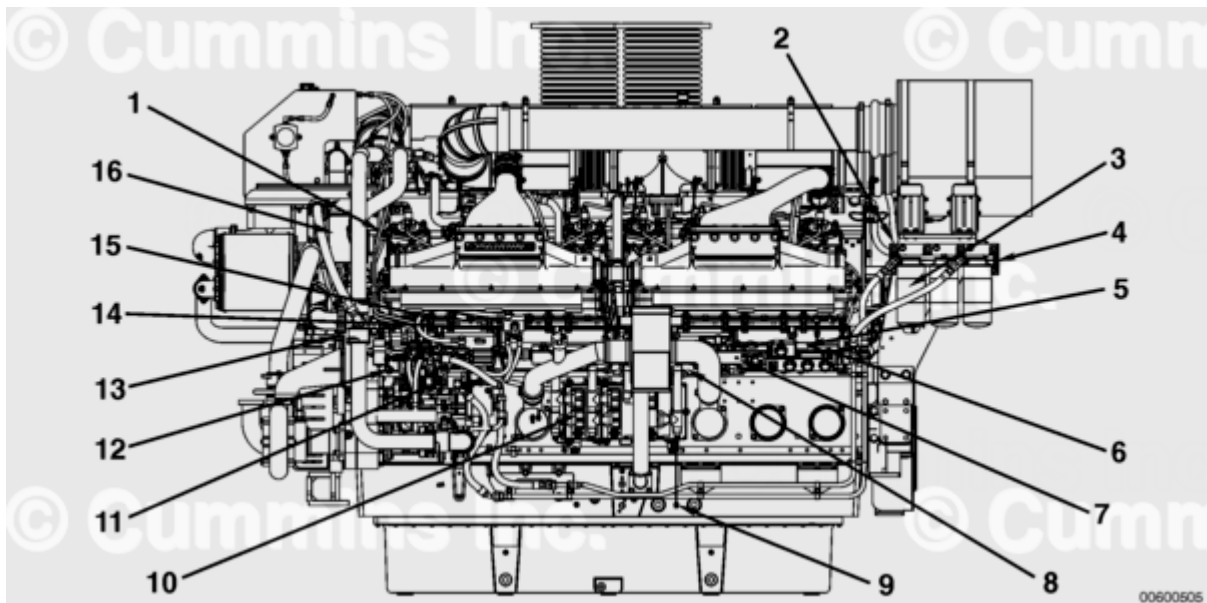
QSK60 Marine Keel Cooled - with Electronically Actuated Injector - Right View

1. Engine exhaust gas stack temperature sensor, right manifold
2. Engine coolant pressure sensor
3. Lubricating oil temperature sensor
4. Intake manifold pressure sensor (rear right bank)
5. Intake manifold temperature sensor (rear right bank).



QS60 Marine Keel Cooled - with Electronically Actuated Injector - Top View

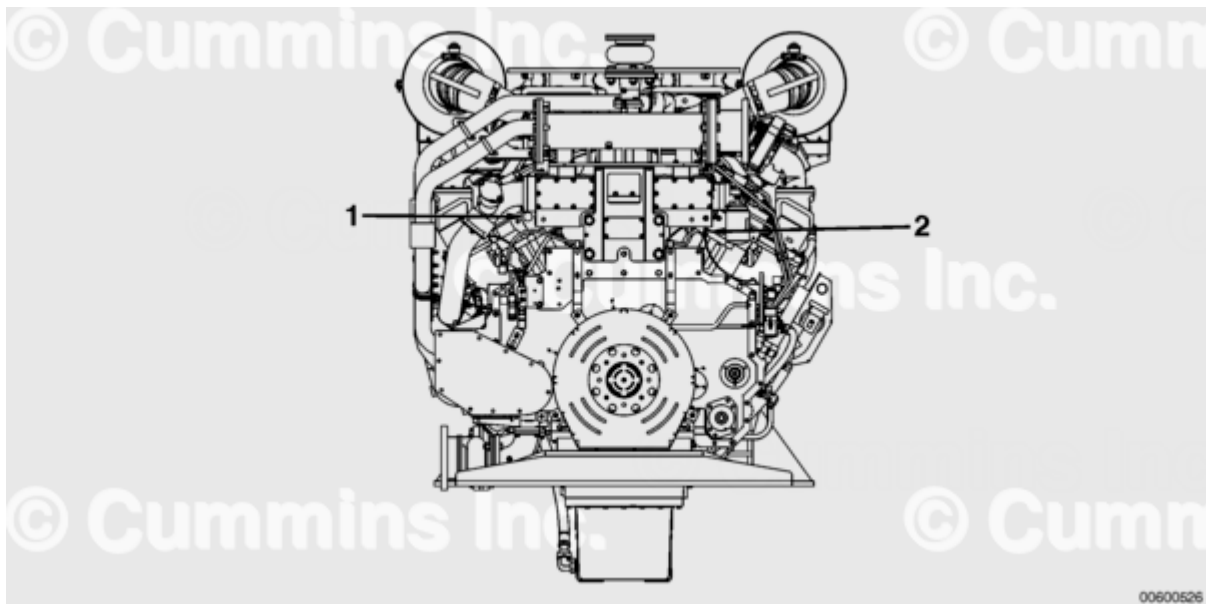
1. Exhaust gas temperature sensor (one per cylinder, located in the exhaust port outlet of the cylinder head)
2. Injector solenoids (one per cylinder, located on top of the rocker housing cover).



QS60 Marine - with Electronically Actuated Injector - Sensor Locations

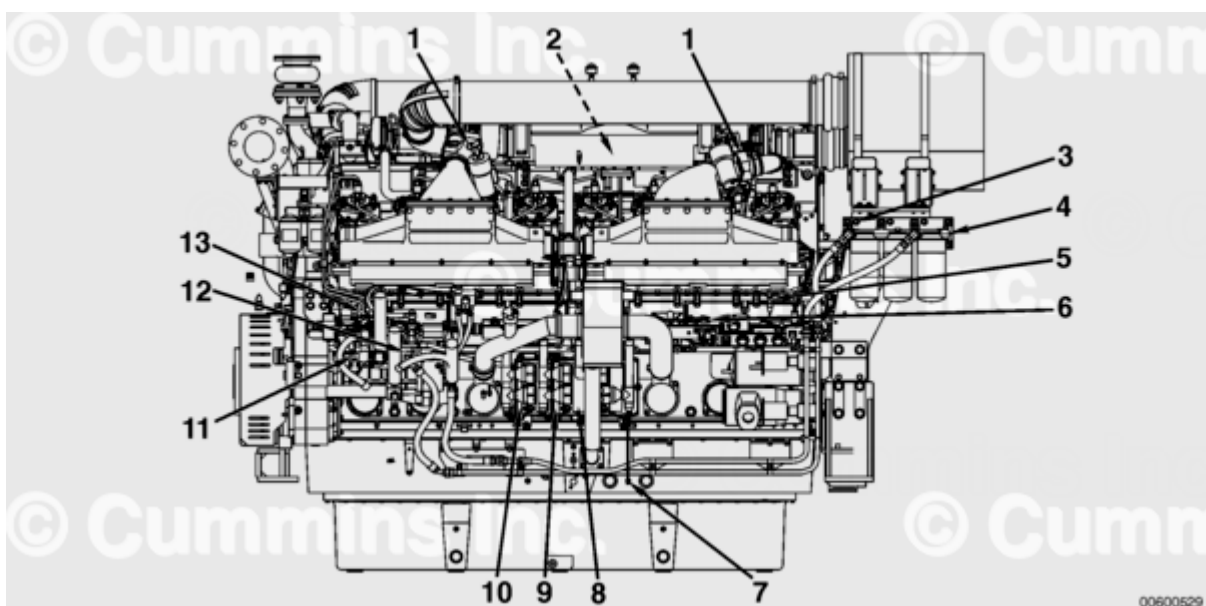
1. Exhaust gas temperature sensors (8 on left bank)
2. Engine fuel temperature sensor 1
3. Stage two fuel filters
4. Fuel delivery pressure sensor
5. Intake manifold 3 temperature sensor
6. Service port
7. Engine oil rifle pressure 1 sensor
8. Barometric pressure sensor
9. Engine oil temperature sensor 1

10. Electronic control modules (ECMs)
11. Fuel pump pressurizing assembly 1
12. Injector metering rail 1 pressure sensor
13. Engine camshaft speed/position sensor
14. Intake manifold 1 pressure sensor
15. Intake manifold 1 temperature sensor
16. Engine coolant temperature 1 sensor.



QSK60 Offshore Drill Rig (Heat Shields Removed) - Front View

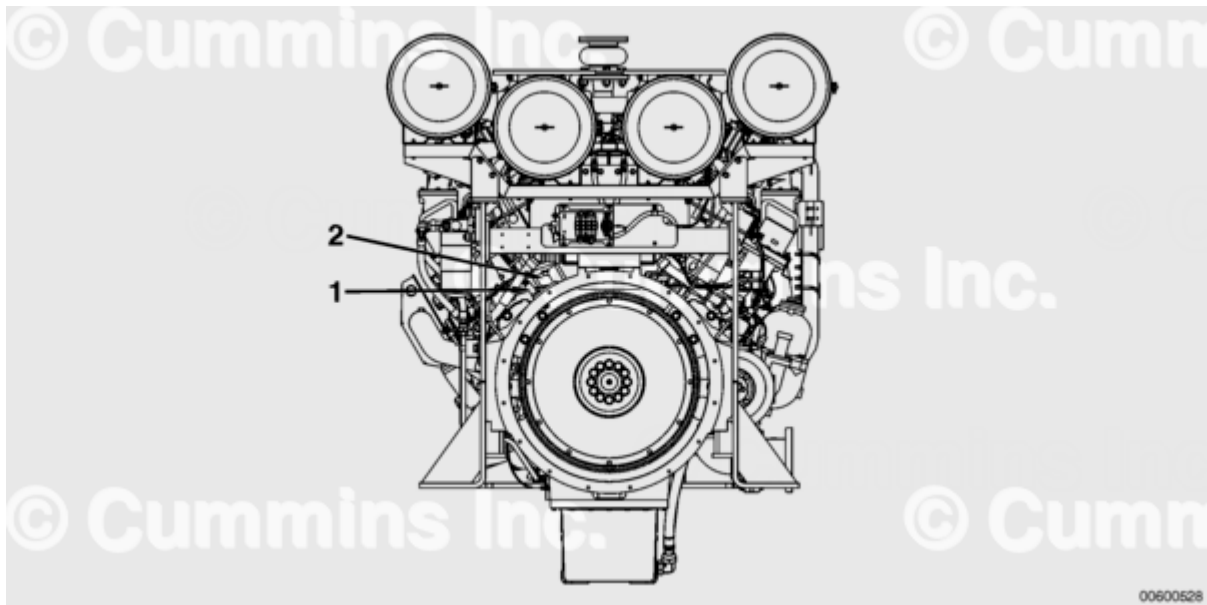
1. Engine coolant pressure sensors (located on right side of engine)
2. Engine coolant temperature sensor.



QSK60 Offshore Drill Rig (Heat Shields Removed) - Left View

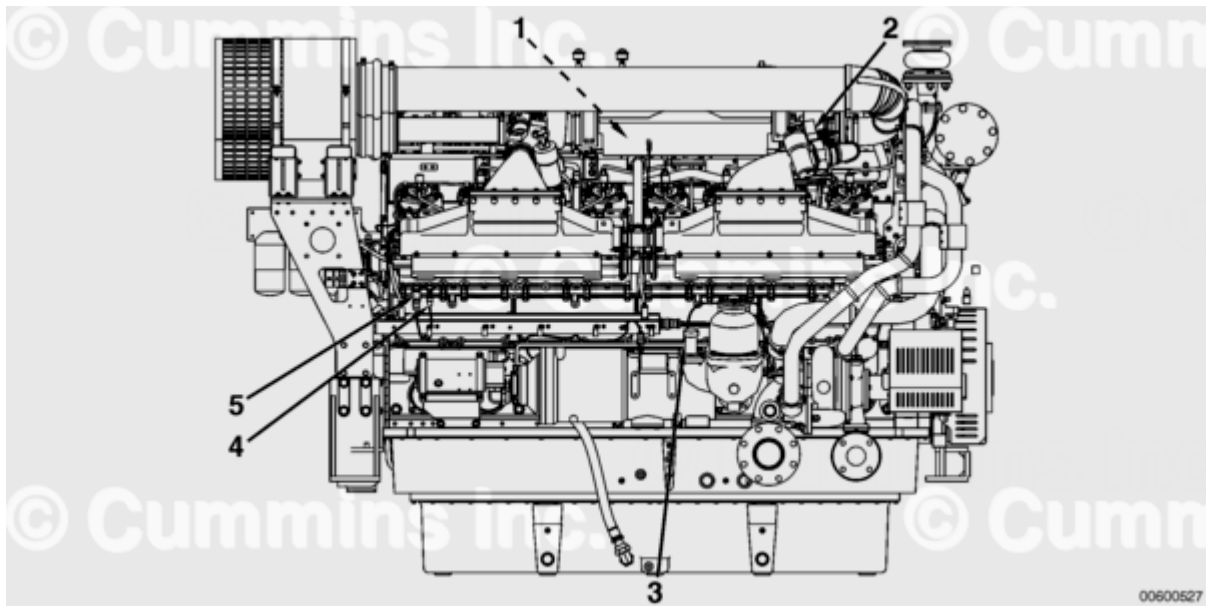
1. Air shutoff valves
2. Engine exhaust gas stack temperature sensor (behind heat shield), left manifold

3. Fuel temperature sensor
4. Fuel supply pressure sensor
5. Intake manifold temperature sensor (one each quadrant)
6. Ambient air pressure sensor
7. Lubricating oil temperature sensor
8. Electronic control module, ECM 1 (source address 144, secondary 2)
9. Electronic control module, ECM 2 (source address 01, secondary 1)
10. Electronic control module, ECM 3 (source address 00, primary)
11. Camshaft position sensor
12. Injector metering rail pressure sensor
13. Intake manifold pressure sensor.



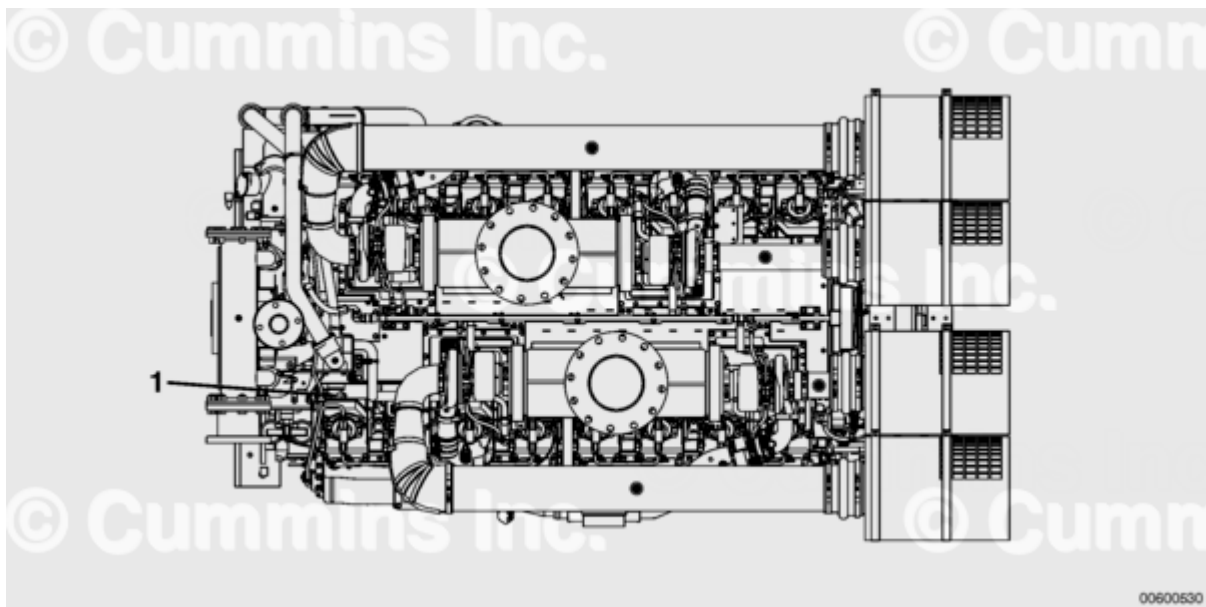
QSK60 Offshore Drill Rig (Heat Shields Removed) - Rear View

1. Engine speed sensor
2. Lubricating oil pressure sensor.



QSK60 Offshore Drill Rig (Heat Shields Removed) - Right View

1. Engine exhaust gas stack temperature sensor (behind heat shield), right manifold
2. Air shutoff valve plunger switch
3. Lubricating oil pre-filter pressure sensor
4. Intake manifold temperature sensor (one in each quadrant)
5. Intake manifold pressure sensor.



QSK60 Offshore Drill Rig (Heat Shields Removed) - Top View

1. Exhaust gas temperature sensors (one per cylinder, located in exhaust outlet in the cylinder head).

